

DIALOGUES, STRATEGIES, AND INTUITIONISTIC PROVABILITY

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0. Introduction

Strategies for dialogues are particular mathematical objects which can serve as *proofs* in systems of logic. They were introduced by P. Lorenzen [11] with the proposal to give an autonomous foundation for intuitionistic logic based on the notion of *dialogue*; such a foundation has been presented recently in Felscher [3]. As is the case with every new methodological device, there arises then the purely mathematical question whether strategies for dialogues actually *do* produce the same results which are obtained by the more familiar methods of proof. It is the purpose of this article to prove an equivalence theorem, saying that every strategy for (certain types of) dialogues can, by a well defined algorithm, be transformed into a proof in Gentzen's calculus LJ and vice versa. On the way to establish this theorem, I shall come across two side results of independent interest.

The basic definitions concerning dialogues and strategies will be presented in § 1, and there then I shall also review the attempts to prove an equivalence theorem existing in the literature. In § 2, I shall introduce several more special types of dialogues and shall prove the corresponding extension theorems, saying that a strategy for a more special type of dialogue can be extended to a strategy for a more general type. Here then the principal extension theorem in 2.2 can be formulated for abstract dialogues or games and is not restricted to applications in logic. In § 3, I shall transform Gentzen's calculus LJ into a calculus of intuitionistic tableaux, and I shall prove that every such tableau can be brought into a certain *irreducible* or normal form; in the case of propositional logic, treated in the Appendix 3.4, this can be applied in order to construct certain irreducible forms of proofs in a sequent calculus. In the final § 4 then, I shall construct transformations between a very special type of strategies and certain irreducible intuitionistic tableaux.

With the exception of § 2.2, the results of this article were contained in the preliminary version Felscher [2]. I am indebted to Messrs. J. Schulte Mönting

and H. Volger for patient discussions when developing the concepts required for the proofs presented here.

1. Basic concepts

Throughout this article, I shall consider a first-order language, built with (one sort of) variables $x, y, \dots$ and terms t ; formulas shall be constructed from atomic formulas with the propositional connectives $\wedge, \vee, \rightarrow, \neg$ and the quantifiers $\forall, \exists$; I shall also consider *special symbols* $\wedge_1, \wedge_2, \vee, \exists$ as independent objects. By an *expression* I understand either a term or a formula or a special symbol. I introduce two further symbols P and Q ; taking two new (and disjoint) copies of the set of expressions, I form for every expression e two new expressions Pe and Qe , the *P-signed* and the *Q-signed version* of the expression e . When appropriate, I shall use X, Y as variables for P, Q , assuming $X \neq Y$.

A natural number is assumed to be the set of all smaller natural numbers. A sequence always shall be a function defined on either a natural number or on the set ω of all natural numbers.

A *tree* S shall be a partially ordered set of elements called *nodes* with the following properties: there exists a *largest* element e_s (the top node), and for every node e the number $\|e\|$ of nodes f such that $e \leq f < e_s$ is *finite*; every node except e_s has exactly one *upper neighbour* but may have arbitrarily many *lower neighbours*. If e is a node with more than one lower neighbour we say that S *ramifies at* e . A *path* in S is a linearly ordered subset of nodes which, together with each of its elements e , contains all the *preceding* nodes f with $e \leq f$; for every node e , let W_e be the path with e as its smallest element. A *branch* is a path which is maximal, i.e. which cannot be extended downwards. For every path A there exists a unique sequence α_A which is an order-reversing bijection, mapping either a natural number or all of ω onto A ; thus $\|\alpha_A(i)\| = i$ holds for every node $\alpha_A(i)$ in A .

Argumentation forms

There are six logical operators: the four propositional connectives and the two quantifiers. Every *composite*, i.e. non-atomic formula determines uniquely its *leading* operator. Every logical operator determines an *argumentation form*. This is a relation, assigning

to an *assertion*, namely every composite signed formula Xu with this particular leading operator,

one or several Y -signed expressions, called *attacks upon* Xu ,

and, unless u is of the form $\neg w$, to every attack one or several X -signed formulas, called *answers* (or *defenses*) to the attack.

The argumentation forms are defined as follows:

- $\wedge$: assertion: $Xw_1 \wedge w_2$,
two attacks: $Y \wedge_1, Y \wedge_2$,
one answer to each attack: Xw_1, Xw_2 .
- $\vee$: assertion: $Xw_1 \vee w_2$,
one attack: $Y \vee$,
two answers: Xw_1, Xw_2 .
- $\rightarrow$: assertion: $Xw_1 \rightarrow w_2$,
one attack: Yw_1 ,
one answer: Xw_2 .
- $\neg$: assertion: $X\neg w$,
one attack: Yw ,
no answer.
- $\forall$: assertion: $X\forall xw$,
one attack for every term t : Yt ,
one answer to each attack: $Xw(t)$.
- $\exists$: assertion: $X\exists xw$,
one attack: $Y\exists$,
one answer for each term t : $Xw(t)$.

Here $w(t)$ is the substitution instance obtained from w if the term t is substituted for the variable x .

Dialogues and Strategies

Let A and D be two new symbols (standing for attack and defense). Let δ be a sequence of signed expressions, and let η be a function defined on the positive members n of $\text{def}(\delta)$ such that $\eta(n)$ is a pair $[m, Z]$ in which m is an element of n and Z is either A or D . The pair δ, η is a *dialogue* if the following properties are satisfied:

- (D00) $\delta(n)$ is P -signed if n is even and Q -signed if n is odd; $\delta(0)$ is a composite formula.
- (D01) If $\eta(n) = [m, A]$, then $\delta(m)$ is a composite formula and $\delta(n)$ is attack upon $\delta(m)$ according to the appropriate argumentation form.
- (D02) If $\eta(p) = [n, D]$ then $\eta(n) = [m, A]$ and $\delta(p)$ is answer to the attack $\delta(n)$ according to the appropriate argumentation form.

The signed formulas occurring as values of δ are called the *assertions* of the dialogue, and the remaining values of δ are *symbolic attacks*; assertions and symbolic attacks together are the *statements* of the dialogue. The numbers in $\text{def}(\delta)$ are called the *positions*, or *places* or *moves*, of the dialogue. The formula

$\delta(0)$ is the *initial* assertion, and if $\delta(0)$ is Pu , then the dialogue is said to be a *dialogue for the formula u*.

If n is a positive position then $\delta \upharpoonright n$, $\eta \upharpoonright n$ shall be the *restriction* of δ , η to arguments j such that $j < n$; in that case, δ , η is an *extension* of $\delta \upharpoonright n$, $\eta \upharpoonright n$, and $\delta \upharpoonright (n+1)$, $\eta \upharpoonright (n+1)$ is an *immediate extension* of $\delta \upharpoonright n$, $\eta \upharpoonright n$.

I shall presently introduce additional properties, called *dialogue rules*, which will have the effect to restrict the possibilities to extend a dialogue after a certain position has been reached. Let H be a class of dialogues defined by such rules; a dialogue belonging to H is said to be *won by P* if it is finite, ends with an even position, and if H does not contain an extension of this dialogue. Assume also that H is closed under the formation of restrictions.

Let S be a tree, let δ be a function defined on all nodes of S , and let η be a function defined on all nodes different from e_s . The triplet S , δ , η is an *H-strategy for a formula u* if it has the following properties:

- (S0) For every branch A of S : the pair $\delta \cdot \alpha_A$, $\eta \cdot \alpha_A$ is an H -dialogue for u won by P .
- (S1) For every node e of S : if $\|e\|$ is odd, then S does not ramify at e .
- (S2) For every node e of S : if $\|e\|$ is even, then e has as many lower neighbours e' in S as there are different immediate extensions in H of the dialogue lying on the path W_e , and these immediate extensions are realized by δ , η on the paths $W_{e'}$.

D-Dialogues and E-dialogues

Let $\delta(n)$ be an attack in a dialogue δ , η and let k be a position such that $n \leq k$. The attack $\delta(n)$ is said to be *open at* the position k if there is no position p such that $n < p \leq k$ and $\eta(p) = [n, D]$ (i.e. $\delta(p)$ answers $\delta(n)$ in the dialogue). It follows that an attack upon a formula $X \neg w$ remains open at all later places.

A *D-dialogue* is a dialogue δ , η satisfying the following rules:

- (D10) P may assert an atomic formula only after it has been asserted by Q before: if $\delta(n) = Pa$ and a is atomic, then there exists m such that $m < n$ and $\delta(m) = Qa$.
- (D11) If p is an X -position and if at $p-1$ there are several open attacks made by Y , then only the *latest* of them may be answered at p : if $\eta(p) = [n, D]$ and if $n < j < p$, $j - n \equiv 0 \pmod{2}$, $\eta(j) = [i, A]$, then there exists q such that $j < q < p$, $\eta(q) = [j, D]$.
- (D12) An attack may be answered at most once: for every n there exists at most one p such that $\eta(p) = [n, D]$.
- (D13) A p -assertion may be attacked at most once: if m is even, then there exists at most one n such that $\eta(n) = [m, A]$.

An *E*-dialogue is a *D*-dialogue satisfying the rule

- (E) *Q* can react only upon the immediately preceding *P*-statement: if *n* in $\text{def}(\delta)$ is odd, then $\eta(n) = [n - 1, Z]$, $Z = A$ or $Z = D$.

Obviously, (E) implies (D13) and, for odd *p* or *n*, also (D11), (D12). I now can formulate the

Equivalence Theorem. *There exist (recursive) algorithms which, for every formula u , transform a proof in Gentzen's calculus LJ (for intuitionistic logic) of the sequent $\Rightarrow u$ into a D-strategy for u and vice versa.*

More precisely, this is the equivalence theorem for *D*-dialogues. An analogous equivalence theorem for *E*-dialogues will actually be an auxiliary result in our proof.

Argumentation forms and strategies for dialogues can be used in order to develop a philosophical foundation for intuitionistic logic; in this development the rules defining *D*-dialogues have a well-motivated meaning, cf. Felscher [3].

Literature on equivalence theorems

D-dialogues were introduced (in a somewhat different form) in Lorenz [9] in an attempt to prove the equivalence theorem; this attempt was repeated in Lorenz [10]. Some parts of Lorenz [9] were corrected in Stegmüller [18], and in Kindt [6] certain claims of Lorenz' were refuted while others were actually proved. Kindt's refutations were acknowledged in footnote no. 12 of Lorenz [10] where it is said that a correction of the erroneous statements in Lorenz [9] would require "ein paar detaillierte technische Vorbereitungen" (cf. a similar remark in footnote no. 16); unfortunately, these detailed technical preparations never have been presented and the gaps in Lorenz' attempt still seem to be open. In the collection Lorenzen–Lorenz [14] the article Lorenz [10] is simply reprinted together with its footnotes; the part of Lorenz [9] to which footnote no. 12 refers has been omitted altogether.

A first correct proof of an equivalence theorems appears to have been given in Kindt [7]; the dialogues studied there, however, are *not D*-dialogues but employ instead of (D11) a different rule. Following the methods introduced by Lorenz, Kindt's work is presented in an (extremely abstract) frame taken from the theory of games. While *D*-dialogues were discussed also in Lorenzen [12, 13] and in Kamlah–Lorenzen [5], a new start was made with Lorenzen–Schwemmer [15], leading to the use of *E*-dialogues. In Lorenzen–Schwemmer [15, p. 59 and p. 71], it is observed that *E*-strategies give rise to a calculus of "Dialogstellungen" which (if some errors concerning the use of quantifiers are corrected) may be transformed into a calculus of Beth-tableaux; thus it appears that provability by *E*-strategies has been shown to imply intuitionistic provability. A new, and simpler, approach to an equivalence theorem for *D*-dialogues was proposed by

Haas [4], and it may be possible that the gaps contained in this attempt can actually be filled. An attempt at an equivalence theorem for E -dialogues is made in Mayer [16], and Krabbe [8] announces another attempt in forthcoming work by Barth and Krabbe.

With the exception maybe of the work of Kindt and Stegmüller, the literature on dialogues suffers quite generally from ambiguous definitions and incomplete, if not erroneous proofs.

2. Strategies

The proof of the equivalence theorem consists of three steps, of which this paragraph contains the first: I shall establish algorithms transforming D -strategies into certain new objects, called *formal E-strategies*.

2.1. Skeletons and formal strategies

The problems to be discussed in this section arise from the presence of quantifiers. The section could be omitted if we were to study only propositional logic.

Let e be a node of an H -strategy S, δ, η which carries as $\delta(e)$ either a formula $P\forall xw$ or an attack $P\exists$ upon a formula $Q\exists xw$. In that case, the tree S has an infinite ramification at e , since every term t determines a lower neighbour of e , carrying either an attack Qt or an answer $Qw(t)$. Strategies, therefore, will in general be infinite objects, and it is my first aim to replace them by finite objects, their *skeletons*.

A position n in a dialogue δ, η is called *critical* if $\delta(n)$ is either $P\forall xw$ or an attack $P\exists$ upon a formula $Q\exists xw$; similarly, the corresponding nodes of a strategy are called critical. A dialogue is said to be *free* if it has the following properties:

An attack $\delta(m)$ upon a formula $P\forall xw$ is of the form Qy where y is a variable not occurring free in any expression $\delta(h)$ with $h < m$.

An answer $\delta(m)$ to an attack $P\exists$ upon a formula $Q\exists xw$ is of the form $Qw(y)$ where y is a variable not occurring free in any expression $\delta(h)$ with $h < m$.

As is usual, the variable y will be called the *eigenvariable* in these situations; that a variable does not occur free in a symbolic expression shall, of course, mean that it does not occur there at all.

An H -skeleton S, δ, η is a triplet with the same properties as an H -strategy, except that now:

In (S0) the H -dialogues carried by the branches A shall also be free.

In (S2) every critical node e has *only one* lower neighbour carrying an attack upon $\delta(e)$ respectively an answer to $\delta(e)$ (although there may be further

lower neighbours of e which realize different possibilities of Q 's, not related to these attacks or answers).

Every H -strategy contains various H -skeletons; a canonical choice is e.g. that in which the eigenvariables are chosen to be the first ones (with respect to a fixed enumeration) which satisfy the eigenvariable conditions.

If H is D or E then every H -skeleton S, δ, η can be extended to an H -strategy. For let E be the set of all nodes of S which would become critical in a strategy; let e be maximal in E . Denote as e_y the lower neighbour of e which carries Qy or $Qw(y)$ with the eigenvariable y ; denote by S_y, δ_y, η_y the triplet consisting of the subtree with top node e_y and the restrictions of the functions δ, η . Let t be a term; define for every node k of S_y the expression $\delta_t(k)$ as the result of the substitution of t for y in $\delta(k)$ (including, if necessary, a change of bound variables) with the additional proviso that, when proceeding downwards, every eigenvariable z , belonging to a node k in E , is changed such that it *remains* an eigenvariable (i.e. does not occur in t either). Define then, for every term t , a separate copy S_t, δ_t, η_t of S_y, δ_y, η_y with a top node e_t ; inserting all these S_t, δ_t, η_t below the node e , we obtain a triplet S^e, δ^e, η^e which at e satisfies the full condition (S2). Moreover, this construction may be performed for all maximal nodes e of E *simultaneously*, resulting in a triplet S^s, δ^s, η^s . The tree S^s now is certainly infinite and contains for every non-maximal node f of E infinitely many copies; each copy f^s , however, satisfies $\|f^s\| = \|f\|$. As the tree S was finite, there is a finite upper bound to all these numbers; hence a finite number of repetitions of the construction of S^s, δ^s, η^s will result in an H -strategy.

It follows from these observations that we now may study the finite D -skeletons and E -skeletons instead of the infinite strategies. In the remainder of this section I shall restrict myself to E -skeletons. Consider now the example of a formula $\exists xa \rightarrow \exists xa$ where a is atomic; there are two E -skeletons for this formula, viz.

0. $P\exists xa \rightarrow \exists xa$		0. $P\exists xa \rightarrow \exists xa$	
1. $Q\exists xa$	$[0, A]$	1. $Q\exists xa$	$[0, A]$
2. $P\exists$	$[1, A]$	2. $P\exists xa$	$[1, D]$
3. $Qa(y)$	$[2, D]$	3. $Q\exists$	$[2, A]$
4. $P\exists xa$	$[1, D]$	4. $P\exists$	$[1, A]$
5. $Q\exists$	$[4, A]$	5. $Qa(y)$	$[4, D]$
6. $Pa(y)$	$[5, D]$	6. $Pa(y)$	$[3, D]$

In the right skeleton, the attack at 3 is answered at 6 with the substitution term y ; this answer must be delayed because the choice of the substitution term depends on the eigenvariable y appearing at 5. There are no phenomena of an analogous type in, say, the sequent calculus, and I shall prove next that the set of formulas u for which there exist E -skeletons remains unchanged if we exclude skeletons showing this type of behaviour. (It also seems that in Lorenzen–Schwemmer [15] and in Haas [4], where an informal use of E -skeletons is made, the possibility that such skeletons occur has been overlooked.)

In order to obtain clear definitions, I introduce new, so-called *formal* argumentation forms for quantifiers:

- $P\forall$: assertion: $P\forall xw$,
 one attack for every eigenvariable $y: Qy$,
 one answer to each attack: $Pw(y)$.
- $Q\forall$: assertion: $Q\forall xw$,
 one attack for every term $t: Pt$,
 one answer to each attack: $Qw(t)$.
- $P\exists$: assertion: $P\exists xw$,
 one attack for every term $t: Qt$,
 one answer to each attack: $Pw(t)$.
- $Q\exists$: assertion: $Q\exists xw$,
 one attack: $P\exists$,
 one answer for every eigenvariable $y: Qw(y)$.

I define a *formal E-dialogue* in exactly the same way in which I defined a free *E-dialogue*, employing conditions on eigenvariables at the positions indicated, only that now the formal argumentation forms are used for quantified formulas. I define a *formal E-strategy* in exactly the same way in which I defined an *E-skeleton*, but now with formal dialogues instead of free ones *and* with the following changes in (S2):

For every node e : if $\|e\|$ is even then, on the lower neighbours of e , there is only one possibility for Q taken into account for

- making an attack QY (case $P\forall$),
- answering an attack $P\exists$ (case $Q\exists$),
- making an attack Qt (case $P\exists$).

A formal *E-strategy*, therefore, is almost the same as an *E-skeleton*, the only difference being that, instead of attacks $Q\exists$, there now are attacks Qt carrying the substitution term t and that, nevertheless, the eigenvariable conditions are satisfied (i.e. eigenvariables chosen after the attack but possibly before the answer still have to respect t). The adjective *formal* refers to this counter-intuitive announcement of t already through Q .

Every formal *E-strategy* can be transformed into an *E-skeleton*: we simply replace all attacks Qt in $P\exists$ -argumentations by $Q\exists$; eigenvariable conditions respecting t will hold a fortiori if t disappears. The converse transformation is much less obvious, still:

Every E-skeleton can be transformed into a formal E-strategy. In the first part of the proof, let us consider a fixed branch of an *E-skeleton* and let δ, η be the *E-dialogue* lying on this branch. In order to transform it into a formal *E-dialogue*, we have to consider all positions i such that $\delta(i)$ is a formula $P\exists xw$

which is attacked; this attack then necessarily takes place at $i + 1$. I shall construct a transformation which has the effect that all these attacks are *formalized*, i.e. that the attack $Q\exists$ is replaced by Qt , with the appropriate substitution term t , *without violating* eigenvariable conditions. All attacks which are not answered in δ, η can be trivially formalized by choosing t as a sufficiently new variable; all attacks $\delta(i+1)$ which are answered immediately by $\delta(i+2) = Pw(t)$ can also be formalized immediately. Consider now an attack $\delta(i+1)$ which is answered by $\delta(i+k) = Pw(t)$ with $k > 2$; then $i > 0$ holds since $\delta(i+2)$ must refer to a position preceding i . Assume further that all attacks in $P\exists$ -argumentations which precede $\delta(i+1)$ have already been formalized (which is, for instance, the case if $\delta(i+1)$ is the *first* such attack not answered immediately).

I now perform an *exchange construction* in that I (1) remove both $\delta(i), \delta(i+1)$ from their position, (2) insert these expressions as immediately preceding $\delta(i+k)$, and (3) formalize the attack which now is at the position $i+k-1$. To this end, I define sequences δ', η' as follows:

$$\begin{aligned} \text{def}(\delta') &= \text{dcf}(\delta), & \text{def}(\eta') &= \text{def}(\eta), \\ \delta'(n) &= \delta(n), & \eta'(n) &= \eta(n) \quad \text{for } n < i, \\ \delta'(i+n) &= \delta(i+n+2), & \eta'(i+n) &= \eta(i+n+2) \quad \text{for } 0 \leq n < k-2, \\ \delta'(i+k-2) &= \delta(i), & \eta'(i+k-2) &= \eta(i), \\ \delta'(i+k-1) &= Qt, & \eta'(i+k-1) &= [i+k-2, A] \end{aligned}$$

and, finally, $\delta'(n) = \delta(n)$ for $n \geq i+k$ and $\eta'(n)$ accordingly. It follows from $i > 0$ that the initial formula $\delta(0)$ is not affected by this exchange; since both $\delta(i), \delta(i+1)$ are exchanged, the alternation between P - and Q -expressions remains preserved. Since $\delta(i)$ may be attacked only once and since $\delta(i+1)$ may be answered only once, the exchange does not disturb references made from other positions. It now is obvious that δ', η' is a dialogue and that the dialogue rules different from (D11) remain in effect. As for (D11), the only new situation is the assertion of $\delta(i)$ as $\delta'(i+k-2)$ if, in δ, η , the formula $\delta(i)$ was a defense. But then also $\delta(i+k)$ was a defense and, therefore, every attack made at a place $i+n+2$, $0 \leq n < i+k-2$, in δ, η must have been answered before $i+k$. Consequently, in δ', η' every attack made at a place $i+n$, $0 \leq n < i+k-2$, will have been answered before $i+k-2$.

Beginning with the first attack in δ, η which was not answered immediately, we thus may proceed downwards and, after a finite number of applications of the exchange construction, arrive at a formal E -dialogue. It should be observed that the exchange construction, applied to $\delta(i), \delta(i+1)$, may have the effect that an attack, namely $\delta(i-1)$, which was answered immediately in δ, η , becomes separated from its answer in δ', η' . In that case, however, $\delta(i-1)$ was formalized already, i.e. of the form Qs with the substitution term s occurring in the answer $\delta(i) = Pw'(s)$. Transferring down $Pw'(s)$ to the position $i+k-2$ in δ', η' does, therefore, not enlarge the possibilities for the choice of eigenvariables if such have

to be chosen at positions $i + n$, $0 \leq n < k - 2$, of δ' , η' : the term s remains present above these positions in $\delta'(i - 1) = \delta(i - 1)$.

There remains, as the second part of the proof, the transformation of an entire E -skeleton S , δ , η . Again, we have to achieve that all attacks in $P\exists$ -argumentations become formalized, and this is unproblematical for attacks which are either not answered or answered immediately. Consider now an attack which is answered, but not immediately so, and assume that all attacks occurring above it have been formalized already. Consider all branches A of S which contain this attack together with an answer, and let δ_A , η_A be the dialogue lying on A . All these A contain the node carrying the attack which, therefore, in all their dialogues has the same position, say $i + 1$; hence they all coincide at least up to $i + 2$. The respective answers, however, occur as $\delta_A(i + k(A))$ with various numbers $k(A)$, $k(A) > 2$, and if A_1 , A_2 are two such branches then A_2 will ramify from A_1 at an even position $i + n$, $2 \leq n < i + k(A_1) - 1$, $n < i + k(A_2) - 1$. We now perform the exchange construction with $\delta_A(i)$, $\delta_A(i + 1)$ *simultaneously* for all these branches A , obtaining transformed branches A' with dialogues δ'_A , η'_A . Assume now that there is a branch B of S ramifying from one, and then all, A at the position i (meaning that $\delta_A(i)$ itself was an attack which could be answered by Q); let b be the first node of B not in one, and then all, A . Then the subtree with top node b is cut out and isomorphic copies of it are implanted below every $\delta_A(i + k(A)) = \delta'_A(i + k(A))$ on A' . Finally, all branches C , which are different from the A and the branches B through b , are left unchanged and remain in their positions — the only difference being that those C which ramified from some A at $i + 2 + n$, $0 \leq n < i + k(A) - 3$, now will ramify from A' at $i + n$. In this manner, the attack under consideration becomes formalized with respect to all its answers. Repeating this process a finite number of times (and beginning with the maximal not immediately answered attacks), we obtain a formal E -strategy.

2.2. The extension of strategies

The aim of this section is to prove the

Extension Lemma. *Every E -skeleton can be embedded into a D -skeleton (for the same formula).*

This is a special case of a general extension theorem which can be formulated under the following assumptions.

There are two disjoint sets L_0 , L_1 the elements of which are called (abstract) expressions. Every expression V in L_i , $i = 0, 1$, determines the subset $A(V)$ of L_{1-i} of attacks upon V , and every W in $A(V)$ determines the subset $D(V, W)$ of L_i of answers to W ; if V is in L_0 , then $A(V)$ is finite, and if V is in L_1 , then every set $D(V, W)$ is finite. Dialogues and strategies now are defined as before, with L_0 , L_1 taking the rôle of P -expressions and of Q -expressions; a dialogue is won if it is finite, ends with an even position and cannot be continued. E -dialogues and

D -dialogues are defined as before, but with (D10) omitted. For every positive integer k , a D_k -dialogue is defined with

(D13_k) An even position may be attacked at most k times

instead of (D13); thus the D_1 -dialogues are exactly the D -dialogues. I shall prove the

Extension Theorem. *For every k , every E -strategy can be embedded into a D_k -strategy.*

The concrete situation, developed in the last section, now can be *interpreted* in this abstract frame. If V is $P\forall xw$, then we *identify* the various concrete attacks Qy with a (eigen) variable y such that there is *only one* attack upon V ; if V is $Q\exists xw$ and W is the attack $P\exists$, then we *identify* the various concrete answers $Qw(y)$ with a (eigen) variable y such that there is *only one* answer to $P\exists$. Considering only free concrete dialogues, we then obtain abstract E - and D_1 -strategies as interpretations of E - and D -skeletons. The extension lemma then will follow from the extension theorem if it can be shown that the D_1 - and D_k -strategies constructed there from a concrete E -skeleton also are D_1 - and D_k -skeletons, i.e. that they carry concrete free dialogues satisfying also (D10). This fact will be secured below during the construction of Q -mixtures.

During the following proof of the extension theorem, all concepts shall refer to the abstract frame unless the concrete case is explicitly mentioned. I shall speak of *even* and *odd* statements in a dialogue, meaning that they occur at odd or even positions; I shall speak of even or odd nodes e of a tree, meaning that $\|e\|$ is even or odd.

The idea of the proof is quite simple. Consider an even node a of an E -strategy, and let V be the statement carried by a . All attacks upon V , as well as all answers to V if V happens to be an attack, are realized at the lower neighbours of a . The D_k -rules, being less restrictive than the E -rules, may permit to carry out these attacks and answers also at some lower nodes, e.g. following some even node b such that $b < a$. I then shall implant below b new branches by copying the realizations which were present already below a ; in this manner, the new possibilities should become realized also below b . There now arise two difficulties which have to be treated with care. The first one is this: the implantation leads to a piecewise composition of dialogues, and it has to be made certain that the dialogue rules, particularly (D11) and (D12), remain preserved under this composition. The second difficulty comes from the fact that the implantation, while realizing one of the latent possibilities, creates new nodes in the implanted copies and so may multiply the other latent possibilities. Obviously, the implantation has to be performed repeatedly, and it has to be made certain that this happens in such a particular order than the entire process comes to an end after finitely many steps.

The following lemma will be the key when settling the first difficulty. I consider a dialogue satisfying (D11) and at least the even half of (D12): odd attacks may be answered at most once.

Lemma 1. *If an odd attack at q is open at an even h , $q < h$, then there exists an even attack at g , $q < g \leq h$, which is open at h .*

For let C be the set of all j such that $q < j \leq h$; the subset C_0 of even j 's contains one more element than the subset C_1 of odd j 's. Since $\delta(q)$ is open at h , it follows from (D11) that no odd attack made before q is answered in C_0 . It follows from the even half of (D12) that the set C_{0D} of answers in C_0 can be mapped injectively into the set C_{1A} of attacks in C_1 . Also, the set C_{0A} of those attacks in C_0 which are answered *within* C can be mapped injectively into the set C_{1D} of answers in C_1 (assigning always the *first* answer). Consequently, $C_{0D} \cup C_{0A}$ can be mapped injectively into C_1 and, therefore, must be a *proper* subset of C_0 . An element g of C_0 , not belonging to this subset, carries an attack not answered in C , i.e. open at h .

Under the same assumptions, we obtain

Lemma 2. *If an even attack at k is answered at p , $k < p$, then no odd attack made at q , $k < q < p$, is open at p .*

For assume that there exists q such that $k < q < p$; since $\delta(p)$ answers $\delta(k)$, it follows from (D11) that $p - 1$ must carry an answer. Consequently, if q carries an odd attack open at p then this attack is open also at $p - 1$. It follows from Lemma 1 that there exists an even attack at g , $p < g \leq p - 1$, which is open at $p - 1$, and this contradicts (D11).

Analogous Lemmas 1', 2', obtained by interchanging "even" and "odd", hold for dialogues satisfying (D11), (D12). For such dialogues, therefore, the implication expressed in (D11) holds in the stronger form which is obtained if the assumption $j - n \equiv 0 \pmod{2}$ (i.e. "made by Y ") is omitted.

I shall now describe the first basic construction: the Q -mixture of two dialogues. Let δ^1, η^1 and δ^2, η^2 be two finite dialogues, both ending at even positions, which coincide at least up to some position x_0 common to both $\text{def}(\delta^1)$, $\text{def}(\delta^2)$. Let $\xi = \langle x_i \mid i < m \rangle$, $\psi = \langle y_i \mid i < m' \rangle$ be two sequences of even numbers x_i in $\text{def}(\delta^1)$ and y_i in $\text{def}(\delta^2)$ such that $x_0 = y_0$, $x_i < x_{i+1}$ for $i \leq m - 2$, $y_i < y_{i+1}$ for $i \leq m' - 2$; assume that x_{m-1} is the last element of $\text{def}(\delta^1)$ and $y_{m'-1}$ is the last element of $\text{def}(\delta^2)$ and that either (case 1) $m' = m$ or (case 2) $m' = m + 1$. Define

$$(x_i, y_j) = y_j + (x_i - x_0) = x_i + (y_j - y_0) \quad \text{for } i < m, j < m';$$

let h be the largest of these numbers, and let k be a number such that $(x_0, y_0) < k \leq h$. I define

$$\begin{aligned} \text{if } (x_i, y_i) < k \leq (x_i, y_{i+1}) & \quad \text{then } k \text{ is vertical, } \|k\| = i, \quad k_2 = k - (x_i - x_0), \\ \text{if } (x_i, y_{i+1}) < k \leq (x_{i+1}, y_{i+1}) & \quad \text{then } k \text{ is horizontal, } \|k\| = i + 1, \quad k_1 = k - (y_{i+1} - y_0). \end{aligned}$$

For numbers z in $\text{def}(\delta^1)$ or $\text{def}(\delta^2)$ I define $|z|_1$ to be the smallest i such that $z \leq x_i$ and $|z|_2$ to be the smallest j such that $z \leq y_j$. If z is in $\text{def}(\delta^1)$ and $|z|_1 > 0$, then I define

$$z_{13} = z + (y_{|z|_1} - y_0)$$

which is horizontal and satisfies $(z_{13})_1 = z$; if z is in $\text{def}(\delta^2)$ and $|z|_2 > 0$, then I define

$$z_{23} = z + (x_{|z|_2-1} - x_0)$$

which is vertical and satisfies $(z_{23})_2 = z$. On the other hand, if k is vertical, then $y_{\|k\|} < k_2 \leq y_{\|k\|+1}$ and $(k_2)_{23} = k$, and if k is horizontal, then $x_{\|k\|-1} < k_1 \leq x_{\|k\|}$ and $(k_1)_{13} = k$.

I now define a dialogue δ^3, η^3 by $\text{def}(\delta^3) = h + 1$, $\text{def}(\eta^3) = (h + 1) - \{0\}$ and

$$\begin{aligned} \delta^3(k) &= \delta^1(k), & \eta^3(k) &= \eta^1(k) & \text{for } k \leq (x_0, y_0), \\ \delta^3(k) &= \delta^2(k_2), & \eta^3(k) &= [n_{23}, Z] & \text{for } k \text{ vertical}, & \eta^2(k_2) &= [n, Z], \\ \delta^3(k) &= \delta^1(k_1), & \eta^3(k) &= [n_{13}, Z] & \text{for } k \text{ horizontal}, & \eta^1(k_1) &= [n, Z]. \end{aligned}$$

In the *concrete case*, it is clear that (D10) holds for δ^3, η^3 if it holds for δ^1, η^1 and δ^2, η^2 . However, if δ^1, η^1 and δ^2, η^2 are *free* dialogues, then an eigenvariable, appearing in δ^1, η^1 at z with $x_0 < z$, may also occur in expressions $\delta^2(z')$ with $y_0 < z'$ and may, therefore, cease to be an eigenvariable in δ^3, η^3 —this situation occurs in particular if both dialogues coincide also beyond x_0 . Making use of sufficiently new variables, we can rename eigenvariables such as these in both δ^1, η^1 and δ^2, η^2 , and the dialogue δ^3, η^3 constructed after the renamings then will be free.

Assume now that both δ^1, η^1 and δ^2, η^2 are D_k -dialogues. A position k in $\text{def}(\delta^3)$ with $(x_0, y_0) < k$ is an *obstacle* if it carries either an attack violating (D13_k) or an answer violating (D11) or (D12). If there exists an *odd* obstacle, then let h_0 be the first one and define δ^4, η^4 as the restriction $\delta^3 \upharpoonright h_0, \eta^3 \upharpoonright h_0$; otherwise, let δ^4, η^4 be δ^3, η^3 . I shall say that δ^4, η^4 is the *Q-mixture* of δ^1, η^1 and δ^2, η^2 induced by ξ, ψ .

Lemma 3. δ^4, η^4 is a D_k -dialogue.

Here (D13_k) certainly holds since it concerns odd obstacles only. The proof now consists in showing that, in δ^3, η^3 , every even obstacle for (D11) or (D12) is preceded by an odd obstacle. Observe that it follows from the definition of δ^3, η^3 that no horizontal position can answer a vertical attack and that no vertical position can answer a horizontal attack. It will be convenient to write, for positions k with $k \leq (x_0, y_0)$, simply $k_1 = k_2 = k$ and $\|k\| = |k|$.

Let d be a horizontal position answering an attack v . Then d_1 answers the attack v_1 in δ^1, η^1 . If $\|v\| = \|d\|$, then d can neither be a (D11)- nor a (D12)-obstacle; assume from now on $\|v\| < \|d\|$. Consider first the case that d is a

(D11)-obstacle: there exists an odd attack w with $v < w < d$ and open at d . If w would be non-vertical, then w_1 would be an odd attack open at d_1 , violating (D11) in δ^1, η^1 ; thus w must be vertical. It follows from $v < w < d$ that $v_1 < x_{\|w\|} < d_1$; it follows from Lemma 1 that there exists an even attack g in δ^1, η^1 with $v_1 < g \leq x_{\|w\|}$ and open at $x_{\|w\|}$; it follows from Lemma 2' that g is answered in δ^1, η^1 at some u with $x_{\|w\|} < u < d_1$; it follows from the definitions that u_{13} answers the even attack g_{13} in δ^3, η^3 . Since w is vertical, it follows from $w < d$ that $w \leq (x_{\|d\|-1}, y_{\|d\|})$; since w is odd, we have even $w < (x_{\|d\|-1}, y_{\|d\|})$ and w is open at this position. Consequently, w_2 is open at $y_{\|d\|}$. It follows from $x_{\|w\|} < u < d_1$ that $\|w\| < |u_1| \leq \|d\|$; abbreviating $|u_1| = q$ we find $\|w\| + 1 \leq q \leq \|d\|$, and it follows from $w_2 \leq y_{\|w\|+1}$ that $w_2 \leq y_q \leq y_{\|d\|}$. Thus $w_2 < y_q$ since w_2 is odd, and w_2 is open at y_q . It follows from Lemma 1 that there exists an even attack f in δ^2, η^2 with $w_2 < f \leq y_q$ and open at y_q . Hence f_{23} is an even attack in δ^3, η^3 with $f_{23} \leq (y_q)_{23} = (x_{q-1}, y_q)$ and open at this position. But $q = |u_1| = \|u_{13}\|$ implies $(x_{q-1}, y_q) < u_{13} \leq (x_q, y_q)$, and thus the vertical f_{23} remains open at u_{13} . Finally, $g_{13} < f_{23}$ since $g \leq x_{\|w\|}$ implies $g_{13} < w$ and since $w_2 < f$ implies $w = (w_2) < f_{23}$. Thus u_{13} is in odd obstacle preceding d .

Consider now the case that d is a (D12)-obstacle: there exists an earlier answer g to v . If g would be non-vertical, then again (D12) would be violated already in δ^1, η^1 . Thus g must be vertical, and as d was horizontal, this implies $v < (x_0, y_0)$; also, g_2 answers $v_2 = v$ in δ^2, η^2 . Since v is open at (x_0, y_0) it follows from Lemma 1 that there exists an even attack e in both δ^1, η^1 and δ^2, η^2 with $v < e \leq (x_0, y_0)$ and open at (x_0, y_0) . It follows from Lemma 2' that this attack must be answered in δ^1, η^1 at some u with $u < d_1$ and in δ^2, η^2 at some w with $w < g_2$. Thus u_{13} and w_{23} both are answers to e in δ^3, η^3 , and they are different since the one is horizontal and the other is vertical. Consequently, the larger of u_{13}, w_{23} is an odd obstacle in δ^3, η^3 , and u_{13} precedes d because of $u < d_1$, and w_{23} precedes even g because of $w < g_2$. — This concludes the study of horizontal answers. The case of vertical answers is perfectly symmetrical.

For the following definitions I consider an arbitrary D_k -dialogue δ, η . Let t, s be two even positions in $\text{def}(\delta)$ such that $s < t$; I shall consider pairs $[V, s]$ such that V is an abstract expression which is an attack upon $\delta(s)$ or an answer to $\delta(s)$, and I shall define what it means that $[V, s]$ is possible for t . There are situations in which V may be an attack as well as an answer (e.g., in the concrete case, if $\delta(s)$ is an attack $Y(v \rightarrow w)$ upon $X((v \rightarrow w) \rightarrow v)$ and V is Xv); thus I shall distinguish the attack $[V_A, s]$ from the answer $[V_D, s]$. Similarly, the rule (D13_k) may permit that an attack may be performed repeatedly; thus I shall distinguish between the copies $[V_0, s], \dots, [V_{k-1}, s]$ of the attack $[V, f]$. Still, I shall mostly write $[V, f]$ alone with the understanding, however, that V may carry labels. I define that the pair $[V, s]$ is possible for t if the dialogue $\delta \upharpoonright (t+1), \eta \upharpoonright (t+1)$ permits an immediate extension to a D_k -dialogue such that (1) the (unlabelled) expression V appears at the new position $t+1$ and refers to $\delta(s)$, (2) if V is labelled with A or D , then the reference is the appropriate one, (3) if $[V, s]$ is numerically labelled as

$[V_i, s]$, then there are already exactly i attacks V upon $\delta(f)$ in $\delta \uparrow (t+1)$, $n \uparrow (t+1)$. It follows that, if $[V_i, s]$ is possible for t , then no other $[V_j, s]$, $j \neq i$, is possible for t . I denote by $\pi(t)$ the set of all pairs which are possible for t . Finally, the following notational convention will be convenient: if $[V, s]$ is a labelled pair $[V_i, s]$, then $[V', s]$ shall be a pair $[V_j, s]$ with $j \leq i$ where the precise value of j is determined by the context; if $[V, s]$ is not numerically labelled, then $[V', s]$ is $[V, s]$.

I now return to the Q -mixture δ^4, η^4 with the notations introduced before Lemma 3. For each of the dialogues $\delta^1, \eta^1, \delta^2, \eta^2, \delta^4, \eta^4$ there now are functions π^1, π^2, π^4 .

Lemma 4. *Let k be horizontal and $[V, s] \in \pi^4(k)$. If s is horizontal, then $[V, s_1] \in \pi^1(k_1)$; if s is vertical, then $[V, s_2] \in \pi^2(y_{\|k\|})$; if $s < (x_0, y_0)$, then $[V', s] \in \pi^1(k_1) \cup \pi^2(y_{\|k\|})$.*

Let k be vertical and $[V, s] \in \pi^4(k)$. If s is horizontal, then $[V, s_1] \in \pi^1(x_{\|k\|})$; if s is vertical, then $[V, s_2] \in \pi^2(k_2)$; if $s < (x_0, y_0)$, then $[V', s] \in \pi^1(x_{\|k\|}) \cup \pi^2(k_2)$.

Let k be horizontal, $[V, s] \in \pi^4(k)$, and assume first that s is horizontal. If V is an attack, then every realization r of V between s_1 and k_1 determines a realization r_{13} of V between s and k ; thus $[V', s_1] \in \pi^1(k_1)$. But no attacks upon $\delta^4(s)$ are realized at vertical positions, and so $[V, s_1] \in \pi^1(k_1)$. If V is an answer, then no r with $s_1 < r < k_1$ can carry an answer preceding that at k_1 because such answer would result in an earlier answer r_{13} ; if r with $s_1 < r < k_1$ would carry an (even) attack open at k_1 , then also r_{13} would be an attack, and it would be open at k since it could not be answered at a vertical position. Thus $[V, s_1] \in \pi^1(k_1)$ holds in any case. Assume next that s is vertical. If V is an attack, then every realization r of V between s_2 and $y_{\|k\|}$ determines a realization r_{23} between s and k , and since now there are no horizontal realizations, it follows that $[V, s_2] \in \pi^2(y_{\|k\|})$. If V is an answer, then no r with $s_2 < r < y_{\|k\|}$ can carry an answer V or an attack open at $y_{\|k\|}$; thus $[V, s_2] \in \pi^2(y_{\|k\|})$ holds in any case. Assume, finally, that $s < (x_0, y_0)$ holds. If V is an attack, then we conclude $[V', s] \in \pi^1(k_1)$ and $[V', s] \in \pi^2(y_{\|k\|})$ as before, but now V' may have a smaller index than V . If V is an answer, then every earlier realization r of V between s, k_1 or between $s, y_{\|k\|}$ would produce an earlier realization r_{13} or r_{23} . Also, every attack r made between $(x_0, y_0), k_1$ or between $(x_0, y_0), y_{\|k\|}$ must be answered before k_1 or $y_{\|k\|}$ respectively since otherwise it would produce an attack r_{13} or r_{23} open at k . Every attack r with $s < r \leq (x_0, y_0)$, not answered already before (x_0, y_0) , must be answered between one of the pairs $(x_0, y_0), k_1$ and $(x_0, y_0), y_{\|k\|}$ but not necessarily between both of these pairs; let $\tilde{r}$ be the smallest of such attacks r with one-sided answers only. If $\tilde{r}$ now is answered between $(x_0, y_0), k_1$ then also every other one-sided r must be answered between $(x_0, y_0), k_1$ since (D11) holds for δ^1, η^1 ; if $\tilde{r}$ is answered between $(x_0, y_0), y_{\|k\|}$, then also every other r must be answered there since (D11) holds for δ^2, η^2 . Consequently, depending on $\tilde{r}$, $[V, s]$ is in $\pi^1(k_1)$ or in $\pi^2(y_{\|k\|})$.

This concludes the case of a horizontal k . The case of a vertical k is perfectly symmetrical.

The developments presented so far were concerned with the combinatorics of single dialogues; I shall come now to constructions with strategies. The embedding of an E -strategy into a D_k -strategy will be constructed in a series of successive extensions. The objects arising during this process will be at least as rich as are E -strategies but, maybe, not quite as rich as are D_k -strategies: they are triplets having the properties of a D_k -strategy with the following two differences: their branches carry D_k -dialogues which, although possibly not won by P , are extensions of E -dialogues won by P ; their ramifications at even nodes take into account all possibilities offered by the E -rules but, maybe, do not take into account all possibilities offered by the D_k -rules. I shall call such triplets *prepre-strategies*, and if all their branches end with an even node I shall call them *prestrategies*. If S, δ, η is a prestrategy and e, f are nodes with $f \geq e$, then I shall say that a pair $[V, f]$ is possible for e if $[V, \|f\|]$ is possible for $\|e\|$ in the dialogue lying on W_e ; again, V may be labelled as in the case of dialogues. I denote by $\pi(e)$ the set of all pairs which are possible for e . I say that a pair $[V, f]$, which is possible for e , is *realized for e* if there exists a lower neighbour e' of e in S such that the dialogue on $W_{e'}$ has the properties (1), (2), (3) guaranteeing the possibility of $[V, f]$ for e . I shall say that a pair $[V, f]$ is *latent for e* if it is possible but not realized for e . I denote by $\rho(e)$ the set of all pairs realized and by $\lambda(e)$ the set of all pairs latent for e . It follows that, if $[V_i, f]$ is latent for e , then *no* $[V_j, f]$, $j < k$, is realized for e .

Let S be a tree and let e be a node in S ; I shall write $S(e)$ for the set of all h such that $e \geq h$ and $MS(e)$ for the set of all h such that $e > h$. If $e', e'', \dots$ are lower neighbours of e , then any union of (one or several) sets $S(e')$, $S(e'')$, $\dots$ shall be called an *e -bush*. — I shall describe now an important auxiliary construction: the implantation. Consider a prestrategy S, δ, η and even nodes a, d, g in S such that $a \geq d$, $a > g$. Let φ be an order-preserving injection, defined on the set of all e with $a > e \geq d$, and satisfying

$$a > \varphi(e) \geq g, \quad \delta\varphi(e) = \delta(e), \quad \eta\varphi(e) = [\varphi(f), Z] \quad \text{if } \eta(e) = [f, Z]$$

where I write $z = \varphi(z)$ for $z \geq a$. Such φ exists in particular if $d = a$ or $d \geq g$. Let M be a d -bush and let M' be an isomorphic copy of M , built on a set of new nodes not belonging to S ; let σ be the isomorphism from M' onto M . Let S' be the tree obtained from S by implanting M' below g . Then the functions δ, η can be extended to M' , defining $\delta(m) = \delta\sigma(m)$, $\eta(m) = [\sigma^{-1}(n), Z]$ if $\eta\sigma(m) = [n, Z]$ for every m in M' ; here $\sigma^{-1}(n)$ shall be $\varphi(n)$ for $n \geq d$. Every branch A' through g leading into M' is the copy of a branch A through d leading into M , and it follows from the definitions that A' now carries a dialogue. Cutting off A' at the first obstacle, we obtain a path A'' , but we *cannot* be certain that A'' ends with an even node. Let S'' be the tree obtained from S' by (1) cutting down all the A' to paths A'' and (2) cutting off those A'' at g which have the property that the lower

neighbour of g in A'' carries an expression which, with the same reference, occurs also at one of the lower neighbours of g in S ; let M'' be the set of nodes from M' remaining in S'' . If m is even in M' , then the possibilities offered by the E -rules for $\sigma(m)$ are realized in M and, therefore, are copied into M' ; these copied realizations at lower neighbours of m , referring immediately to m , cannot be obstacles and thus remain in M'' if m belongs to M'' . Consequently, S'', δ, η is a preprestrategy which I shall call the φ -implantation at g of the reduced copy M'' of M . Every branch of S'' leading into M'' is of the form A'' for some (not uniquely determined) A' copied from a branch A of S leading into M . — More generally, implantations such as these will be constructed not only with respect to *one* node g but *simultaneously* with respect to an entire family of such nodes. As for the *concrete case*, the only additional observations required is that the eigenvariables may have to be renamed during these constructions.

Let S, δ, η be a prestrategy, let a, b be even nodes with $a > b$, and let a' be a lower neighbour of a (and the case $a' > b$ is *not* excluded!). The *basic embedding* S^*, δ^*, η^* of S, δ, η for a', b shall be a prestrategy with the following property: if $\eta(a') = [f, Z]$ and if $[\delta(a'), f]$ is latent at b , then b obtains in S^* a new lower neighbour realizing $[\delta(a'), f]$, and *all* possibilities to continue from this copy of a' , repeating this copying process as long as possible, are realized in S^* . *In principle* this construction of S^* is quite clear: we have to take all dialogues lying on branches A through a' , and also all dialogues lying on branches B through b , and then we have to form the Q -mixtures of these A -dialogues with the B -dialogues induced by all possible finite sequences. These new dialogues then have to be implanted on new branches below b , and this in such a manner that any two of the new dialogues which coincide up to a certain (necessarily even) position are realized on branches which ramify precisely at this position. It is only this last task, the partial unification of branches carrying partially coinciding Q -mixtures, which requires a more systematic organization: I shall construct S^*, δ^*, η^* as the last member of a finite sequence of successive embeddings S_i, δ_i, η_i of S, δ, η .

Define S_0, δ_0, η_0 as S, δ, η . Define S_1, δ_1, η_1 to be the implantation at b of a reduced copy M'' of $S(a')$; here the nodes a, d, g needed for the implantation are chosen as a, a, b and φ is empty. In order to see that the preprestrategy S_1, δ_1, η_1 is actually a prestrategy, consider a branch A'' in S_1 leading into M'' . We then can find a branch A' leading into the unreduced copy M' of $S(a')$ such that A'' is the result of cutting off A' at the first obstacle; let A be the branch in S copied by A' , and let p be the last (and even) node in A . If δ^1, η^1 is the dialogue on A and if δ^2, η^2 is the dialogue on W_b , then it follows from the definitions that the dialogue on A' is the same as the dialogue δ^3, η^3 which gives rise to the Q -mixture δ^4, η^4 of δ^1, η^1 and δ^2, η^2 induced by the sequences $\langle \|a\|, \|p\| \rangle, \langle \|a\|, \|b\| \rangle$. It follows from Lemma 3 that the first obstacle in δ^3, η^3 is odd; thus A'' carries the Q -mixture δ^4, η^4 and, therefore, ends with an even node. — I now define $c_1 = b$ and write $G(c_1)$ for M'' ; I also define $b_1 = b$ and write σ_1 for the isomorphism from $G(c_1)$ into $S(a')$. Finally, I define σ_0 to be the identity on the set of all e such that $a > e \geq b_1$.

Define S_2, δ_2, η_2 to be obtained from S_1, δ_1, η_1 by simultaneous implantation, containing, for every even c_2 in $G(c_1)$, a reduced copy $G(c_1, c_2)$ of $MS(b_1)$ at c_2 . Here the nodes a, d, g , needed for the implantation of $G(c_1, c_2)$, are chosen as a, b_1, c_2 , and φ is the map σ_0^{-1} (which happens to be the same as σ_0). In order to see that S_2, δ_2, η_2 is a prestrategy, consider a branch B'' leading into $G(c_1, c_2)$ which arises from a branch B' copied from a branch B leading into $MS(b_1)$; let q be the last node in B and define $a_1 = \sigma_1(c_2)$. It then follows that the dialogue on B'' is the Q -mixture of the dialogues on W_{a_1} and on B , induced by $\langle \|a\|, \|a_1\| \rangle, \langle \|a\|, \|b_1\|, \|q\| \rangle$; thus B'' ends with an even node. — Let $\mathbf{F}_2$ be the family of all the sets $G(c_1, c_2)$ and let σ_2 be the union of the isomorphisms embedding the various $G(c_1, c_2)$ into $MS(b_1)$.

I now shall define recursively $S_{n+1}, \delta_{n+1}, \eta_{n+1}$ for $n \geq 2$; in order to do so, there have to be defined already for every $i, 2 \leq i \leq n$,

- (a) a prestrategy S_i, δ_i, η_i ,
- (b) a family $\mathbf{F}_i$ of sets $G(c_1, \dots, c_i)$,
- (c) a map σ_i ,

satisfying the following conditions (1)–(6):

- (1) S_i, δ_i, η_i is an extension of $S_{i-1}, \delta_{i-1}, \eta_{i-1}$.
- (2) The set in $\mathbf{F}_i$ are pairwise disjoint subsets of S_i , and their union is the set of nodes in S_i which are not in S_{i-1} .
- (3) Every set $G(c_1, \dots, c_i)$ in $\mathbf{F}_i$ is uniquely determined by the sequence $\langle c_0, \dots, c_i \rangle$.
- (4) If $i > 2$ and if $G(c_1, \dots, c_i)$ is in $\mathbf{F}_i$, then there is a set $G(c_1, \dots, c_{i-1})$ in $\mathbf{F}_{i-1}$, and c_i is an even node contained in $G(c_1, \dots, c_{i-1})$.
- (5) σ_i is defined on the union of the sets in $\mathbf{F}_i$ and maps every set $G(c_1, \dots, c_i)$ isomorphically onto a subset of $MS(\sigma_{i-2}(c_{i-1}))$. If $\sigma_{i-2}(c_{i-1}) > x > z$ and if z is a value under σ_i of an element in $G(c_1, \dots, c_i)$, then so is x .

Before stating the last property (6), I define for every $G(c_1, \dots, c_i)$ the abbreviations $a_0 = b_0 = a$, $a_j = \sigma_{2j-1}(c_{2j})$ for $0 < 2j \leq i$, and $b_{j+1} = \sigma_{2j}(c_{2j+1})$ for $0 \leq 2j < i$. Observe that a set $G(c_1, \dots, c_i, \dots, c_{i+r})$ defines, for $2j \leq i$ or $2j < i$, the same a_j, b_{j+1} as does $G(c_1, \dots, c_i)$.

- (6) (i) If $i = 2m, m > 0$, then the branches in S_i leading into $G(c_1, \dots, c_i)$ carry precisely those dialogues which are Q -mixtures of the dialogue on W_{a_m} and the dialogues on branches B in S containing b_m , induced by $\langle \|a_0\|, \|a_1\|, \dots, \|a_m\| \rangle, \langle \|b_0\|, \|b_1\|, \dots, \|b_m\|, \|q\| \rangle$ where q is the last node in B .
- (ii) If $i = 2m + 1, m \geq 0$, then the branches in S_i leading into $G(c_1, \dots, c_i)$ carry precisely those dialogues which are Q -mixtures of the dialogues on branches A in S containing a_m , and the dialogue on $W_{b_{m+1}}$, induced by $\langle \|a_0\|, \dots, \|a_m\|, \|p\| \rangle, \langle \|b_0\|, \|b_1\|, \dots, \|b_{m+1}\| \rangle$ where p is the last node in A .

Define S_{n+1} , δ_{n+1} , η_{n+1} to be obtained from S_n , δ_n , η_n by simultaneous implantation, containing, for every $G(c_1, \dots, c_n)$ and for every even c_{n+1} in $G(c_1, \dots, c_n)$, a reduced copy $G(c_1, \dots, c_{n+1})$ of $MS(\sigma_{n-1}(c_n))$ at c_{n+1} . For this implantation of $G(c_1, \dots, c_{n+1})$ the nodes a, d, g are chosen as $a, \sigma_{n-1}(c_n), c_{n+1}$, and the map φ is chosen as the following map φ_γ depending on the sequence $\gamma = \langle c_1, \dots, c_{n-1} \rangle$. If $n+1 = 2m$, $m > 0$, then φ_γ shall be defined on the set of all x with $b_0 > x \geq b_m$ as follows. If $j < m$ then b_{j+1} is the value of the element c_{2j+1} in $G(c_1, \dots, c_{2j})$, and thus σ_{2j} determines a bijection of the set of all y with $c_{2j} > y \geq c_{2j+1}$ onto the set of all x with $b_j > x \geq b_{j+1}$; let $\varphi_\gamma(x)$ be the unique y satisfying $\sigma_{2j}(y) = x$. If $n+1 = 2m+1$, $m > 0$, then φ_γ shall be defined on the set of all x with $a_0 > x \geq a_m$ as follows. If $j \leq m$, then a_j is the value of the element c_{2j} in $G(c_1, \dots, c_{2j-1})$, and thus σ_{2j-1} determines a bijection of the set of all y with $c_{2j-1} > y \geq c_{2j}$ onto the set of all x with $a_{j-1} > x \geq a_j$; let $\varphi_\gamma(x)$ be the unique y satisfying $\sigma_{2j-1}(y) = x$.

Defining $\mathbf{F}_{n+1}$ to be the family of all $G(c_1, \dots, c_{n+1})$ and defining σ_{n+1} to be the union of all the isomorphisms embedding the $G(c_1, \dots, c_{n+1})$ into $MS(\sigma_{n-1}(c_n))$, it is clear that the conditions (1)–(5) are satisfied for every i with $2 \leq i \leq n+1$. I shall prove now (6), and then it follows again that the preprestrategy S_{n+1} , δ_{n+1} , η_{n+1} is a prestrategy. Consider the case $n+1 = 2m+1$ and let A'' be a branch in S_{n+1} leading into $G(c_1, \dots, c_{n+1})$; let A' be a branch leading into the unreduced copy of $MS(\sigma_{n-1}(c_n)) = MS(a_m)$ giving rise to A'' , and let A be a branch through a_m copied by A' . Then A' passes through c_{n+1} , and $W_{c_{n+1}}$ is a path contained in a branch B'' leading into $G(c_1, \dots, c_n)$. Since (6) holds for n , it follows that the dialogue on B'' is the Q -mixture of the dialogue on W_{a_m} and the dialogue on a branch B in S containing b_m , induced by $\langle \|a_0\|, \dots, \|a_m\| \rangle, \langle \|b_0\|, \dots, \|b_m\|, \|q\| \rangle$ where q is the last node in B ; consequently, the dialogue on $W_{c_{n+1}}$ is the Q -mixture of the dialogue on W_{a_m} and the dialogue on $W_{b_{m+1}}$, induced by $\langle \|a_0\|, \dots, \|a_m\| \rangle, \langle \|b_0\|, \dots, \|b_{m+1}\| \rangle$. If we now inspect the definition of A' as the continuation of $W_{c_{n+1}}$ by the σ_{n+1}^{-1} -copy of that part of A which begins below a_m , then we find that the dialogue on A' is the same as the dialogue δ^3, η^3 which gives rise to the Q -mixture δ^4, η^4 of the dialogues on A and on $W_{b_{m+1}}$, induced by $\langle \|a_0\|, \dots, \|a_m\|, \|p\| \rangle, \langle \|b_0\|, \dots, \|b_{m+1}\| \rangle$ where p is the last node of A . It follows from Lemma 3 that the first obstacle in δ^3, η^3 is odd, and thus A'' carries δ^4, η^4 .—The case $n+1 = 2m+2$ is perfectly symmetrical.

This concludes the recursive definition of the sequence of prestrategies S_n , δ_n , η_n . It follows from the definitions that a set $G(c_1, \dots, c_n)$ will be empty if $MS(\sigma_{n-2}(c_{n-1}))$ is empty, i.e. if $\sigma_{n-2}(c_{n-1})$ is minimal in S . Observe now that $a > \sigma_1(c_2) > \sigma_3(c_4) > \dots$ and $b > \sigma_2(c_3) > \sigma_4(c_5) > \dots$, and as these sequences lie on finite branches of S , it follows that they come to an end after finitely many steps. Consequently, a sequence $G(c_1), G(c_1, c_2), \dots$ arrives at an empty $G(c_1, \dots, c_n)$ after finitely many steps. I now use the fact that the tree S of a prestrategy is finite; in the abstract case this may need the use of König's lemma while in the concrete case no choice principles are needed. Consequently, there exists an upper bound n^* for the lengths of all branches in S , and then every

$G(c_1, \dots, c_n)$ with $n \geq n^*$ will be empty. Thus the sequence of prestrategies S_n , δ_n , η_n becomes stationary at least for $n \geq n^*$, and I define S^* , δ^* , η^* to be S_{n^*} , δ_{n^*} , η_{n^*} ; this prestrategy I call the *basic embedding* of S , δ , η for a' , b .

Every new node e of S^* appears below b and hence there must exist a first n such that e appears in S_n but not in S_{n-1} . In that case, e appears in one of the sets $G(c_1, \dots, c_n)$, and thus an even e may serve as a node c_{n+1} . It follows that the study of new, even nodes in S^* amounts to the study of nodes c_n determined by sets $G(c_1, \dots, c_n)$. Also, if e is in $G(c_1, \dots, c_n)$, then I shall write $\sigma(e)$ for $\sigma_n(e)$. Adapting the terminology used in connection with Q -mixtures, I shall call *horizontal* the nodes c_n (determined by $G(c_1, \dots, c_n)$) if n is even, and *vertical* if n is odd; I shall also count as vertical the nodes y in S such that $a > y \geq b$. We then can translate the statements in Lemma 4 as follows:

Let $[V, f]$ be in $\pi^*(c_{2m})$. If f is horizontal, then $[V, \sigma(f)] \in \pi(a_m)$; if f is vertical, then $[V, \sigma(f)] \in \pi(b_m)$; if $f \geq a$, then $[V', f]$ is in $\pi(a_m) \cup \pi(b_m)$.

Let $[V, f]$ be in $\pi^*(c_{2m+1})$. If f is horizontal, then $[V, \sigma(f)] \in \pi(a_m)$; if f is vertical, then $[V, \sigma(f)] \in \pi(b_{m+1})$; if $f \geq a$, then $[V', f]$ is in $\pi(a_m) \cup \pi(b_{m+1})$.

We thus have found a *reduction* of the possibilities at new nodes to related possibilities at nodes in S . Defining σ on S as being the identity, we can write this also as:

If $[V, f] \in \pi^*(c_{n+1})$, then $[V', \sigma(f)] \in \pi(\sigma(c_{n+1})) \cup \pi(\sigma(c_n))$.

While possibilities depend alone on the dialogue and not on the stage of the construction at which they appear, matters are different for realizations and for latencies. If c_{n+1} is in $G(c_1, \dots, c_n)$, then a pair in $\lambda_n(c_{n+1})$ may become realized in $G(c_1, \dots, c_{n+1})$; thus $\rho_n(c_{n+1}) \subseteq \rho_{n+1}(c_{n+1}) = \rho^*(c_{n+1})$ and $\lambda_n(c_{n+1}) \supseteq \lambda_{n+1}(c_{n+1}) = \lambda^*(c_{n+1})$. I now shall prove:

If $[V, f] \in \pi^*(c_{n+1})$ and $[V', \sigma(f)] \in \rho(\sigma(c_{n+1})) \cup \rho(\sigma(c_n))$,
then $[V, f] \in \rho_{n+1}(c_{n+1})$.

Consider the case $[V', \sigma(f)] \in \rho(\sigma(c_{n+1}))$. The set $G(c_1, \dots, c_n)$ is a reduced copy M'' of $MS(\sigma(c_{n-1}))$, the latter set containing $\sigma(c_{n+1})$. The realization of $[V', \sigma(f)]$ at a lower neighbour of $\sigma(c_{n+1})$ is reproduced as a realization of $[V, f]$ at a lower neighbour of c_{n+1} in the unreduced copy M' of $MS(\sigma(c_{n-1}))$. This reproduction cannot be an obstacle since $[V, f]$ is possible at c_{n+1} ; therefore, it is preserved in M'' . Consider now the case $[V', \sigma(f)] \in \rho(\sigma(c_n))$. The set $G(c_1, \dots, c_{n+1})$ is a reduced copy M'' of $MS(\sigma(c_n))$. The realization of $[V', \sigma(f)]$ at a lower neighbour of $\sigma(c_n)$ is reproduced as a realization of $[V, f]$ at a lower neighbour of c_n in the unreduced copy M' of $MS(\sigma(c_n))$. Again, this reproduction cannot be an obstacle and, therefore, it will be preserved in M'' .—We now conclude from the definitions:

Lemma 5. If $[V, f] \in \lambda^*(c_{n+1})$, then $[V', \sigma(f)] \in \lambda(\sigma(c_{n+1})) \cup \lambda(\sigma(c_n))$.

It is this *reduction of latencies* which was the reason to embark upon the construction of S^* , δ^* , η^* .

I now come to the proof of the Extension Theorem. Beginning with an E -strategy, I shall construct a finite sequence of prestrategies which ends with a D_k -strategy, and each member of which is an embedding of the preceding members. A D_k -strategy can be described as a prestrategy such that $\lambda(e)$ is empty for every (even) node e ; a more practical characterization is the following one. For every prestrategy S, δ, η let R be the set of all nodes r such that there exists e and V with $[V, r] \in \lambda(e)$; then S, δ, η is a D_k -strategy if R is empty. I shall construct, for every prestrategy S, δ, η , an embedding prestrategy $S^\S, \delta^\S, \eta^\S$ such that $R^\S$ is a proper subset of R ; the desired sequence of prestrategies then will be obtained by simply iterating this construction a finite number of times.

Consider S, δ, η and let r be minimal in R . I shall define $S^\S, \delta^\S, \eta^\S$ in such a manner that $R^\S$ is a subset of R not containing r . Let $W^0, \dots, W^{m-1}$ be a fixed listing of the finitely many expressions which are attacks upon $\delta(r)$ or answers to $\delta(r)$; expressions distinguished only by labels shall be listed separately and every sequence $V_0, \dots, V_{k-1}$ of numerically labelled attacks shall appear as a block in this list. For every j with $j < m$ let n_j be the number of nodes e in S such that $[W^j, r] \in \lambda(e)$; the sequence $\nu = \langle n_0, \dots, n_{m-1} \rangle$ shall be called the *order of r* in S, δ, η . I shall define an embedding prestrategy S^+, δ^+, η^+ such that R^+ is a subset of R and r , although maybe still in R^+ , has a *lexicographically smaller* order ν^+ in S^+, δ^+, η^+ . Repeating this construction a finite number of times, I arrive at a prestrategy in which r obtains an order consisting of zeros only, and this prestrategy will be $S^\S, \delta^\S, \eta^\S$.

In order to define S^+, δ^+, η^+ , let n_q be the first positive member of ν , and choose b in S such that $[W^q, r] \in \lambda(b)$. I now distinguish two cases, the first one being that W^q is either an answer or the first one in a block of attacks. In that case the E -rules permit a realization of W^q at a lower neighbour r' of r , and I define $a = r, a' = r'$. In the other case, W^q is an attack V_p with $0 < p < k$. Since $[V_p, r]$ is possible at b , there must already be p realizations of the attack V at nodes between r and b ; let a' be the lowest one of these nodes and let a be the upper neighbour of a' — observe that now $r > a > a' > b$. In both cases, I define S^+, δ^+, η^+ to be the basic embedding S^*, δ^*, η^* of S, δ, η for a', b . It remains to be shown that R^* is a subset of R and that the order ν^* of r is smaller than ν .

The set of new nodes in S^* is of the form $S^*(b')$ where b' is the lower neighbour of b with $\sigma(b') = a'$. The expression $\delta^*(b')$ is a copy of $\delta(a')$, and it follows from the choice of a' that W^q is realized at b' ; thus $[W^q, r]$ is in $\rho^*(b)$ and is not in $\lambda^*(b)$. Consequently, no even e in $S^*(b')$ can have $[W^q, r]$ in $\pi^*(e)$ as follows from (D12) if W^q is an answer and from the definition of possible pairs if W^q is an attack; thus no even e in $S^*(b')$ can have $[W^q, r]$ in $\lambda^*(e)$ either. Since b , which contributed to n_q , does *not* contribute to n_q^* , and since none of the new nodes e contributes to n_q^* either, it follows that $n_q^* = n_q - 1$. Consider now any

situation $[V, f] \in \lambda^*(e)$ for some new e ; we know already $[V, f] \neq [W^q, r]$. It follows from Lemma 5 that $[V', \sigma(f)]$ is latent at a node in S . Since r was chosen to be minimal in R , the relation $r > \sigma(f)$ is impossible, and thus the relation $a > \sigma(f)$ is impossible. This implies $\sigma(f) \geq a$ and, therefore, $\sigma(f) = f$, $f \geq a$. Consequently, f is in S and, therefore, R^* is a subset of R . If $f \neq r$, then $[V, f]$ does not contribute to the order ν^* . If $f = r$, then $[V', r]$ is latent at a node in S , and it follows from the definition of the index q that V' is an expression W^s with $q \leq s$. It follows from the definition of V' that then also V is an expression W^s with $q \leq s$, and now $q = s$ is excluded because of $[V, r] \neq [W^q, r]$. Consequently, e may contribute to n_s^* with $q < s$, but ν^* is lexicographically smaller than ν .

2.3. Conclusion

Collecting the results established so far, we arrive at the following conclusions: For each of the following types of objects

- (1) D -strategies
- (2) D -skeletons,
- (3) E -skeletons,
- (4) formal E -strategies,

we have defined algorithms, transforming the objects of the one type into those of the adjoining types and vice versa. The transformations between (1) and (2) and between (3) and (4) were established in 2.1; the transformations between (2) and (3) were established in 2.2.

Employing (D13_k) also for the concrete case, we can define (concrete) D_k -dialogues. The transformations between (1) and (2) remain in effect between D_k -strategies and D_k -skeletons, and transformations between D_k -skeletons and E -skeletons were established in 2.2. Consequently, we may add as a fifth type

- (0) D_k -strategies

to our list. D_k -dialogues have been considered by Lorenz [10]. Looking for further formal generalizations, one may attempt to generalize (D12) by permitting, say, that a P -attack may be answered at most twice; let me call such dialogues $D_{(2)}$ -dialogues for the moment. The example of the formula $(a \vee b) \rightarrow (a \vee b)$ shows that a D -strategy cannot, in general, be extended to a $D_{(2)}$ -strategy *without* permitting also that a Q -attack may be answered twice. This second permission, however, will also have the effect to create strategies for formulas which are *not* provable intuitionistically; such strategies, therefore, could *not* be transformed into D -strategies.

3. Tableaux

This paragraph contains the second step in the proof of the equivalence theorem: I shall establish algorithms transforming proofs in Gentzen's calculus LJ into certain new objects, called (irreducible) *IC-protableaux*.

3.1. Intuitionistic tableaux and protableaux

Gentzen's calculus LJ for intuitionistic logic is well known, cf. e.g. Curry [1, p. 263]. Let LJ' be the calculus differing from LJ in that the *one* rule for $\rightarrow$ -introduction

$$(I\rightarrow) \quad \begin{array}{c} M, a_1 \Rightarrow a_2 \\ M \Rightarrow a_1 \rightarrow a_2 \end{array}$$

is replaced by the *two* rules

$$(I0\rightarrow) \quad \begin{array}{c} M, a_1 \Rightarrow a_1 \rightarrow a_2 \\ M \Rightarrow a_1 \rightarrow a_2 \end{array} \quad (I1\rightarrow) \quad \begin{array}{c} M \Rightarrow a_2 \\ M \Rightarrow a_1 \rightarrow a_2 \end{array}$$

Since the cut rule is admissible for LJ, both these calculi are equivalent: every LJ-proof can be transformed into a LJ'-proof and vice versa. Now a proof in LJ' can be viewed as a co-tree (i.e. an object obtained from a tree by reversing its order) together with a function assigning sequents to its nodes. To every such proof, I shall define a new object, consisting of a tree together with a function assigning signed formulas to its nodes. These new objects shall be called *IC-tableaux*; they are a straightforward generalization of the so-called *analytic tableaux* introduced by Smullyan [17] and used there for an elegant treatment of classical logic.

Consider signed formulas as introduced in Section 1. To every composite signed formula I assign one or several other signed formulas as its *components*:

$Qw_1 \wedge w_2:$	$Qw_1, Qw_2,$	$Pw_1 \wedge w_2:$	$Pw_1, Pw_2,$
$Pw_1 \vee w_2:$	$Pw_1, Pw_2,$	$Qw_1 \vee w_2:$	$Qw_1, Qw_2,$
$Pw_1 \rightarrow w_2:$	$Qw_1, Pw_2,$	$Qw_1 \rightarrow w_2:$	$Pw_1, Qw_2,$
$P\neg w:$	$Qw,$	$Q\neg w:$	$Pw,$
$Q\forall xw:$	$Qw(t) \text{ for each } t,$	$P\forall xw:$	$Pw(y) \text{ for each } y,$
$P\exists xw:$	$Pw(t) \text{ for each } t,$	$Q\exists xw:$	$Qw(y) \text{ for each } y.$

Quantified formulas, therefore, have infinitely many components. The types of formulas listed in the left schema, as well as the last three types listed in the right schema, are called *unramified*; the first three types in the right schema are called *ramified*.

An *intuitionistic tableau* T, t, ε is a triplet consisting of

- (t0) a tree T with the top node e_T ,
- (t1) a function t assigning signed formulas to the nodes of T ,
- (t2) a reference function ε , defined on the non-minimal nodes of T , with values in T and such that $e \leq \varepsilon(e)$ for every e in $\text{def}(\varepsilon)$,

and such that the following conditions are satisfied:

- (T00) If $t\varepsilon(e)$ is a formula of an unramified type, then e has only one lower neighbour e' and $t(e')$ is a component of $t\varepsilon(e)$.

- (T01) If $t\varepsilon(e)$ is a formula of a ramified type, then e has two lower neighbours and t assigns to them the two different components of $t\varepsilon(e)$.
- (T1) If $t\varepsilon(e)$ is a quantified formula of a type from the right schema, then the *eigenvariable* y in the substitution instance $Pw(y)$ or $Qw(y)$ must not occur free in any formula $t(f)$ such that $f \geq e$.
- (T2) If $\varepsilon(e)$ is P -signed, then $\varepsilon(e)$ is the smallest P -signed node in W_e .

Here a node is said to be P -signed (Q -signed) if it carries a P -signed (Q -signed) formula.—A branch of a tableau (i.e. a branch of the underlying tree T) is *intuitionistically closed* if it contains nodes e, f such that

- (c0) there exists an atomic formula a with $t(e) = Qa$, $t(f) = Pa$,
- (c1) if $e < f$, then f is the smallest P -signed node in W_e .

An *intuitionistically closed* or *IC-tableau* is an intuitionistic tableau such that each of its branches is intuitionistically closed. The conditions (T2) and (c1) are characteristic for the intuitionistic case: if they are omitted, we obtain Smullyan's *closed analytic tableaux* for classical logic. A tableau is said to be a *tableau for a formula* u if Pu stands at its top node.

Every IC-tableau for a formula u can be transformed into a LJ'-proof of the sequent $\Rightarrow u$ and vice versa. Starting from an IC-tableau T, t, ε , we may assume that its branches end at those nodes where they become intuitionistically closed for the first time. To every node e of T we assign a sequent $M \Rightarrow v$, abbreviated as $b(e)$, where M is the set of all m such that $Qm = t(f)$ for some f in W_e and where v is the unique formula such that $Pv = t(f)$ with f minimal in W_e . Reversing the order of T , we obtain a co-tree B which we extend to a co-tree B_1 as follows. Assume that e in T has the lower neighbour e' , $t\varepsilon(e) = Q \neg w$, hence $t(e') = Pw$ and $b(e'): M \Rightarrow w$, while $b(e)$ is of the form $M, \neg w \Rightarrow v$. We then insert a node e^* between e' and e in B and also extend b by $b(e^*): M, \neg w \Rightarrow$, meaning that $b(e)$ arises from $b(e^*)$ by an application of *weakening to the right*. It is easy to see that B_1 , together with the extended function b , is an LJ'-proof.

Conversely, consider an LJ'-proof of the sequent $\Rightarrow u$ with a co-tree B ; we may assume that it begins with generalized axioms of the form $M, a \Rightarrow a$ and does not contain applications of *weakening to the left*. We now form a co-tree B_0 by omitting from B all nodes e^* such that the sequent at e^* is premiss to an application of *weakening to the right*. In that case, the sequent at e^* must have been conclusion of an application of a logical rule to premisses appearing at upper neighbours e', e'' of e^* in B , and in B_0 these e', e'' become upper neighbours of the lower neighbour e of e^* in B ; I shall say that, in B_0 , this same logical rule leads from e', e'' to e . I now define inductively the function t on B_0 , beginning with $t(e_B) = Pu$ for the smallest node e_B of B_0 . If e in B_0 has upper neighbours e', e'' then I consider the logical rule leading to e and define $t(e'), t(e'')$ to be the side formulas of this rule appearing in the premisses at e', e'' . Reversing the order of

B_0 , we obtain a tree T , and it is easy to see that, with a suitably defined reference function ε , the triplet T, t, ε is an IC-tableau for u .

IC-tableaux have at least that much in common with (formal) E -strategies that they are trees carrying signed formulas. But the branches of an E -strategy carry dialogues such that, e.g., the P - and Q -expressions alternate, and IC-tableaux will, in general, not have this property. An examination of possible transformations shows that it is convenient to introduce the auxiliary concept of a *protableau*; the explicit technical reasons leading to our choice of definitions will become clear during the later applications. In addition to signed formulas, I shall use now also Q -signed expressions $Q\wedge_1$, $Q\wedge_2$, $Q\vee$ and Qt for every term t ; all variables occurring in t are said to occur free in Qt . I now extend the definition of components:

- $Q\wedge_i$ is a *procomponent* of $Pw_1 \wedge w_2$ belonging to the component Pw_i , $i = 1, 2$,
- $Q\vee$ is a *procomponent* of $Pw_1 \vee w_2$ belonging to each of its components,
- Qt is a *procomponent* of $P\exists xw$ belonging to the component $Pw(t)$,
- Qy is a *procomponent* of $P\forall xw$ belonging to the component $Pw(y)$.

I shall also say that a procomponent is an attack, answered by the components to which it belongs. An *intuitionistic protableau* T, t, ε shall be a triplet with (t_0) , (t_1) , (t_2) , but the function t in (t_1) now may assign also procomponents, and such that the following conditions are satisfied:

- (PT00) If $t\varepsilon(e)$ is a formula of an unramified type, then e has only one lower neighbour e' and $t(e')$ is either a component or a procomponent of $t\varepsilon(e)$. Every answer is preceded by an attack: if $t(e')$ is a component, then there exists g such that $\varepsilon(e) \geq g > e$, $\varepsilon(g) = \varepsilon(e)$, and the lower neighbour g' of g carries the procomponent belonging to $t(e')$.
- (PT01) If $t\varepsilon(e)$ is a formula of a ramified type, but not of the type $Pw_1 \wedge w_2$, then e has two lower neighbours and t assigns to them the two different components of $t\varepsilon(e)$.
- (PT02) If $t\varepsilon(e)$ is $Pw_1 \wedge w_2$, then e has one or two lower neighbours. If e has two lower neighbours, then t assigns to them the two procomponents of $t\varepsilon(e)$. If e has one lower neighbour e' , then $t(e')$ is a component Pw_i , and this answer is preceded by an attack: there exists g such that $\varepsilon(e) \geq g > e$, $\varepsilon(g) = \varepsilon(e)$, the node g has two lower neighbours, and if g' is that one satisfying $g' \geq e$, then $t(g')$ is the procomponent $Q\wedge_i$.
- (PT10) If $t\varepsilon(e)$ is $Q\exists xw$, then the eigenvariable in the substitution instance $Qw(y)$ must not occur free in any expression $t(f)$ such that $f \geq e$.
- (PT11) If $t\varepsilon(e)$ is $P\forall xw$ and if $t(e')$ is the procomponent Qy , then the eigenvariable y must not occur free in any expression $t(f)$ such that $f \geq e$.
- (PT2) If $\varepsilon(e)$ is P -signed, then $\varepsilon(e)$ is the smallest P -signed node in W_e .

(PT3) If $t\varepsilon(e)$ admits procomponents, then every component $t(e')$ is preceded by *exactly one* attack on $t\varepsilon(e)$.

The last condition means that (1) the attack answered by $t(e')$ occurs only once between $\varepsilon(e)$ and e , and (2) no other attack on $t\varepsilon(e)$ (to be answered by another component) occurs between $\varepsilon(e)$ and e . Readers interested in exploring the possibilities of protableaux are invited to show that it suffices to demand (PT3) only for the case $t\varepsilon(e) = P\forall xw$: every protableau satisfying this weaker condition (together with all the others) can be transformed into a protableau satisfying (PT3) — in particular, if only propositional languages are considered then (PT3) can be omitted. — Intuitionistically closed branches and intuitionistically closed or *IC-protableaux* are defined by the same conditions which were used for tableaux.

The following simple constructions will be applied repeatedly. Consider a triplet T, t, ε satisfying at least (t_0) – (t_2) , and let h, e be nodes such that $h \geq e$. If e has one lower neighbour e' , the triplet T', t', ε' obtained by omitting e' and changing references to h is defined as follows. T' is the tree obtained from T by omitting e' , and t' is the restriction of t to T' ; the function ε' is defined by $\varepsilon'(e) = \varepsilon(e')$, $\varepsilon'(f) = h$ for all f in T' such that $\varepsilon(f) = e'$, and $\varepsilon'(f) = \varepsilon(f)$ for all other nodes f . — If e has two lower neighbours e', e'' , the triplet T', t', ε' obtained by omitting e' , proceeding below e' and changing references to h is defined as follows. The tree T' is obtained from T by omitting e' as well as all nodes g such that $e'' \geq g$; the function t' is the restriction of t to T' , and ε' is defined as before.

If q is an expression which is either a component or a procomponent of $t(h)$, the triplet T', t', ε' obtained by inserting q below e and proceeding as before is defined as follows. T' is obtained from T by inserting a new node k such that k is the only lower neighbour of e in T' and the subtree of all g in T with $g < e$ is copied isomorphically into the subtree of all g in T' with $g < k$. The function t' extends t to T' by $t'(k) = q$, the function ε' is defined by $\varepsilon'(e) = h$, $\varepsilon'(k) = \varepsilon(e)$, $\varepsilon'(f) = \varepsilon(f)$ for all other nodes f . — In the same manner I shall define triplets obtained by inserting several new expressions and proceeding in accordance with given directions.

Let T, t, ε be an IC-protableau; the *omission of unanswered attacks* is the following transformation into another IC-protableau T', t', ε' for the same formula. Let G be the set nodes in T carrying attacks which remain unanswered. If g', g'' in G are both lower neighbours of g with $t\varepsilon(g) = Pw_1 \wedge w_2$, then choose one of g', g'' and remove the other from G ; let G_0 be the remaining subset of G . Let g' be in G ; omitting g' and proceeding below g' we obtain an IC-protableau (although, if $t\varepsilon(g) = Pw_1 \wedge w_2$ holds for the upper neighbour g of g' , then the entire subtree descending from the other lower neighbour g'' of g disappears, independently on whether the attack $t(g'')$ was answered or not!). Beginning with the maximal nodes, we thus omit all nodes in G_0 and arrive at T', t', ε' .

The *separation of $Q\forall$ -attacks* is the following transformation into another IC-protableau T', t', ε' for the same formula. Assume that h is a node with

$t(h) = Pw_1 \vee w_2$ and consider nodes e with lower neighbours e' such that $\varepsilon(e) = h$ and $t(e')$ is a component. We know that there exists g such that $h \geq g > e$ and $t(g)$ is the attack $Q\vee$ answered by $t(e')$ — but it may well happen that there exists another pair e_0, e'_0 on another branch such that $g > e_0$ and also $t(e'_0)$ answers the attack $Q\vee$. We now first omit the node g carrying $Q\vee$, and then we insert the attack $Q\vee$ below e proceeding with e' , below e_0 proceeding with e'_0 , and analogously for other pairs e_1, e'_1 in the same situation. Performing this construction for all possible nodes h , we arrive at T', t', ε' .

Every IC-tableau for a formula u can be transformed into an IC-protatableau for the same formula and vice versa. Assume first that T, t, ε is an IC-tableau. If e is in T and $t\varepsilon(e)$ is a formula admitting procomponents, but not $Pw_1 \wedge w_2$, then we insert the procomponent belonging to $t(e')$ below e and proceed as before. If $t\varepsilon(e)$ is $Pw_1 \wedge w_2$ and $t(e') = Pw_1, t(e'') = Pw_2$ holds for the lower neighbours e', e'' of e , then we insert both procomponents below e and proceed with e' below $Q\wedge_1$ and with e'' below $Q\wedge_2$. Performing this construction for every possible e , we obtain an IC-protatableau.

Assume now that T, t, ε is an IC-protatableau, and assume that unanswered attacks have been omitted and that $Q\vee$ -attacks have been separated. Let e be in T with a lower neighbour e' such that $t(e')$ is a component of the formula $t\varepsilon(e)$ which admits procomponents. It follows from (PT3) that there exists a unique g_e carrying the attack answered by $t(e')$; define T', t', ε' by first omitting e' and changing references to g_e and afterwards re-defining $t'(g_e) = t(e')$ — thus the P -formula $t(e')$ is shifted upwards to the node g_e . This shift does not obstruct references made to $t(e')$ (either in order to dissolve this formula or in order to obtain an intuitionistically closed branch) since T does not contain P -signed nodes f such that $g_e \geq f \geq e$. Nor does it obstruct references made to $t\varepsilon(e)$ from some node h : it follows from (PT3), together with the separatedness of $Q\vee$ -attacks, that in such a case either not $g_e \geq h$ or $g_e = g_h$ holds. And neither does it obstruct references made from nodes h with $g_e \geq h$ to P -formulas different from $t\varepsilon(e)$ — no such P -formulas occur on nodes f such that $\varepsilon(e) > f \geq g_e$. Performing this construction of T', t', ε' for all possible nodes e , we then obtain an IC-tableau.

I conclude this section with the description of yet another transformation between IC-protatableaux. In this connection it will be convenient to extend the terminology of attacks and answers to the case of formulas $Pw_1 \rightarrow w_2$: here I shall say that the component Pw_2 , if preceded by Qw_1 , is an answer to the attack Qw_1 . For an IC-protatableau T, t, ε I define that a node e of T is *in use* if one of the following conditions holds:

- (u0) There exists a branch A of T containing e such that $t(e)$ is one of the two atomic formulas, the lower of which intuitionistically closes A for the first time.
- (u1) There exists a node f such that $e = \varepsilon(f)$, and one of the lower neighbours of f is in use.

- (u2) e carries an attack, and there exists an answer to this attack at a node which is in use.

Consider now an arbitrary IC-protableau for a formula u . If k is a node on a branch A such that A becomes intuitionistically closed through $t(k)$ for the first time, then no node below k is in use. Cutting off all branches A below their respective nodes k , we arrive at a finite IC-protableau T', t', ε' such that every minimal node of T' is in use. If T' contains nodes which are not in use, let e be a minimal one, and let T'', t'', ε'' be obtained by omitting e and proceeding below e . Then T'', t'', ε'' is an IC-protableau in which all nodes below the upper neighbour f of e in T' are in use (although $\varepsilon''(f)$ may not be in use!). Repeating this process a finite number of times, we arrive at an IC-protableau for the same formula u in which every node is in use; IC-protableaux having this property shall be called *useful*.

3.2. Irreducible protableaux

The proof of the equivalence theorem requires a construction which transforms IC-protableaux — or IC-tableaux — into E -strategies; this construction will be presented in § 4. In order to be subjected to this construction, an IC-protableau should have the special property that P -formulas occur only in such positions where they are actually put to use — either for immediate dissolution or for intuitionistically closing a branch. In general, of course, the P -formulas will not be in such positions: they will be *reducible*, and reductions will be needed in order to bring them into *irreducible* positions. These reductions will be of two types, depending on the kind of formulas.

Let T, t, ε be an IC-protableau. All formulas of the type $P\neg w$ and $Pw_1 \rightarrow w_2$, as well as the nodes carrying them, are said to be of the *first kind*. A node e of the first kind is *irreducible* if the following conditions hold:

- (r0) $\varepsilon(e) = e$, and if $t(e) = Pw_1 \rightarrow w_2$, then $t(e') = Qw_1$ for the lower neighbour e' of e .
- (r1) If $t(e) = P\neg w$, then $g < e$ implies $\varepsilon(g) \neq e$; if $t(e) = Pw_1 \rightarrow w_2$, then $g < e$, $\varepsilon(g) = e$ implies $t(g') = Pw_2$ for the lower neighbour g' of g .

The node e is *reducible* if it is not irreducible. Assume now that T contains reducible nodes of the first kind and let e be a maximal one. If e violates (r0), then we insert the missing component below e . If e , together with certain nodes g , violates (r1), then we omit the lower neighbours g' of all these g , beginning with the maximal ones among these g . We then obtain an IC-protableau in which e is irreducible and no new reducible nodes have arisen. Repeating this process a finite number of times, we arrive at an IC-protableau in which every node of the first kind is irreducible; such a protableau I shall call *1-irreducible*. If the construction removing unused nodes is applied to a 1-irreducible IC-protableau, the resulting useful IC-protableau remains 1-irreducible.

All P -formulas, as well as the nodes carrying them, which are *not* of the first kind, shall be of the *second kind*. A node e of the second kind is *irreducible* if either $\varepsilon(e) = e$ or if e is minimal in T and carries a P -signed atomic formula; otherwise e is *reducible*.

The reduction process for formulas of the first kind was quite simple since these formulas themselves did not change their positions. Moreover, this reduction works just as well in IC-tableaux and does not require the use of procomponents and protableaux. For formulas of the second kind, however, changes of positions may become necessary; as an example, consider the following IC-tableau together with two IC-protableaux:

$P\forall xw \rightarrow \forall xw$	$P\forall xw \rightarrow \forall xw$	$P\forall xw \rightarrow \forall xw$
$Q\forall xw$	$Q\forall xw$	$Q\forall xw$
$P\forall xw$	$P\forall xw$	$P\forall xw$
$Pw(y)$	Qy	Qy
$Qw(y)$	$Pw(y)$	$Qw(y)$
	$Qw(y)$	$Pw(y)$

If we assume that w is atomic, then the IC-tableau to the left is the *only* IC-tableau for $\forall xw \rightarrow \forall xw$ since the eigenvariable y in $Pw(y)$ must be chosen *before* the substitution term y in $Qw(y)$ is determined. In the first IC-protableau then $Pw(y)$ is reducible, in the second $Pw(y)$ has been shifted into an irreducible position. This change is possible only because the eigenvariable y is kept in a position preceding that of $Qw(y)$ through the use of the procomponent Qy . Situations of this kind are the reason for the introduction of procomponents of quantified formulas and for the use of protableaux; procomponents of propositional formulas are a convenience which, in principle, could be avoided — this will become clear in 3.4.

During the reductions to be described presently, I shall make frequent use of the following fact concerning eigenvariables. Let e be a node of an IC-protableau such that the lower neighbour e' of e carries a procomponent or a component introducing an eigenvariable y , and let z be another variable which, just as it is the case for y , does not occur free in any expression $t(f)$ with $f \geq e$. For nodes h with $h < e$ let $t'(h)$ be the result of substituting z for y in $t(h)$; for all other nodes h define $t'(h) = t(h)$. Then T, t', ε is again an IC-protableau, said to be obtained by a change of eigenvariables from the original IC-protableau T, t, ε .

Consider now a useful, 1-irreducible IC-protableau T, t, ε and let e be reducible of the second kind; I shall define the reduction process *at the node* e . Define

$\Gamma(e)$ to be the smallest set of nodes containing e and such that if d is in $\Gamma(e)$ and $\varepsilon(d) \neq e$, then also those lower neighbours of d are in $\Gamma(e)$ which are Q -signed.

$\Delta(e)$ to be the set obtained from $\Gamma(e)$ by removing e ; since e is in use and reducible, $\Delta(e)$ is not empty and contains only nodes carrying Q -formulas.

The *boundary* of $\Gamma(e)$ to be the set of all d in $\Gamma(e)$ which have one lower neighbour inside $\Gamma(e)$ and another one outside $\Gamma(e)$; these are precisely those d in $\Gamma(e)$ for which $t_\varepsilon(d)$ is a $Q \rightarrow$ -formula.

The *base* of $\Delta(e)$ to be the set of minimal elements of $\Delta(e)$; if d belongs to the base then either (1) $\varepsilon(d) = e$ or (2) d is minimal in T or (3) $t_\varepsilon(d)$ is of the form $Q \neg w$. If a base node d is minimal in T , then $t(d)$ is a Q -signed atomic formula; thus $t(e)$, being the lowest P -signed formula above $t(d)$, must be the corresponding Q -signed atomic formula.

The *essential base nodes* to be those d satisfying (1) or (2).

The node e has an upper neighbour f since e_T is irreducible. Depending on the situation at f , I shall define two different reductions, each of which transforms T, t, ε into another useful, 1-irreducible IC-protatableau T', t', ε' and shifts the formula $t(e)$ into an irreducible position. The new protatableau shall coincide with the old one for all nodes h such that *not* $h < f$; thus it remains to describe how to proceed from f downwards. In each case, I begin by omitting e and inserting the set $\Delta(e)$ immediately below f ; on $\Delta(e)$ the functions t', t shall be the same and, with the exception of the essential base nodes, also $\varepsilon', \varepsilon$ shall coincide; for f we have $\varepsilon'(f) = \varepsilon(e)$. Further, t', ε' shall coincide with t, ε on all those branches which leave $\Gamma(e)$ at a node d which is from the boundary or is an unessential base node: these branches are copied from T, t, ε . At an essential base node r , however, the following changes will be made.

Case 1. Situation: f has only e as a lower neighbour.

Reduction: Below each r , insert $t(e)$ on a new lower neighbour r_e and then proceed as before below r (i.e. $t'(r_e) = t(e)$, $\varepsilon'(r) = \varepsilon(f)$ and, if r is not minimal in T , also $\varepsilon'(r_e) = \varepsilon(r)$).

Case 2. Situation: f has a second lower neighbour g .

Reduction: Below each r , insert both $t(e)$, $t(g)$ on new lower neighbours r_e, r_g of r (i.e. $t'(r_e) = t(e)$, $t'(r_g) = t(g)$, $\varepsilon'(r) = \varepsilon(f)$). Proceed below r_e as before below r ; proceed below r_g as before below g , changing eigenvariables wherever necessary.

In order to verify that T', t', ε' is an IC-protatableau, observe first that the insertion of the Q -formulas from $\Delta(e)$ will not cause obstructions for the new reference function ε' . The shift of $t(e)$ from e to the nodes r_e removes this P -formula from those branches of T which leave $\Gamma(e)$ at a node d which is either from the boundary or an unessential base node. Such a node d has exactly one lower neighbour d' outside of $\Gamma(e)$, and d' is P -signed. Consequently, the formula $t(e)$ was not used on the branches from which it has been removed: neither can a node h with $h \leq d'$ refer to $t(e)$, nor can $t(e)$ be an atomic formula intuitionistically closing a branch through d' . — Due to the presence of procomponents, the shift of $t(e)$ has no effect on eigenvariable conditions. In case 2, however, there

may occur substitution instances on nodes in $\Delta(e)$ above r introducing free variables which were not present in the expressions appearing above f . Eigenvariables introduced on nodes below g may, therefore, not respect these new variables; hence they will have to be changed when the nodes below g are copied into nodes below r_g .—Thus T', t', ε' is an IC-protatableau, and it follows immediately that each of its nodes will be in use if each of the nodes of T, t, ε was so; also, the property of being 1-irreducible remains preserved. Finally, T', t', ε' has the following property:

Assume that in T, t, ε every node h of the second kind, satisfying $h < e$ or (in the case 2) satisfying $h < g$, is irreducible. Then in T', t', ε' every node k of the second kind, satisfying $k < f$, is also irreducible. Nodes h in T such that *not* $h < f$ have the same status with respect to reducibility in both protatableaux.

An IC-protatableau is called *irreducible* if it is useful, 1-irreducible, and if every node of the second kind is also irreducible. We now can prove the

Reduction Theorem. *Every IC-protatableau T, t, ε can be transformed into an irreducible one for the same formula.*

For assume that T, t, ε is already useful and 1-irreducible. Let $M(T, t, \varepsilon)$ be the set of all nodes of the second kind which are reducible, and let $m(T, t, \varepsilon)$ be the number of elements in $M(T, t, \varepsilon)$; assume that $M(T, t, \varepsilon)$ is not empty. Choose e in $M(T, t, \varepsilon)$ such that the number $\|e\|$ is maximal; let T', t', ε' arise from the reduction process at the node e . The node g in T is Q -signed since there are no dissolutions in a protatableau producing two lower neighbours which both are P -signed; thus $\|g\| = \|e\|$ implies that there are no reducible nodes of the second kind below g in T . Consequently, in T', t', ε' there are no reducible nodes k of the second kind such that $k < f$. The status of nodes h such that *not* $h \leq f$ has not been changed; the node f itself cannot have been irreducible of the second kind in T, t, ε since such a node would have lower neighbours which are Q -signed. Thus it follows that the number $m(T', t', \varepsilon)$ is less than $m(T, t, \varepsilon)$. Repeating this reduction a finite number of steps, we arrive at an irreducible IC-protatableau.

3.3. Conclusion

Collecting the results established so far, we arrive at the following conclusions: for each of the following types of objects

- (9) LJ-proofs of sequents $\Rightarrow u$,
- (8) LJ'-proof of sequents $\Rightarrow u$,
- (7) IC-tableaux,
- (6) IC-protatableaux,
- (5) irreducible IC-protatableaux,

we have defined algorithms, transforming the objects of one type into those of the adjoining types and vice versa.

3.4. Appendix: irreducibility in the propositional case

The result in this appendix are not needed for the proof of the equivalence theorem. I shall show that, in the case of a propositional language, every IC-tableau can be transformed into an irreducible one. As a consequence, every LJ'-proof can be transformed into an irreducible normal form.

Consider from now on a propositional language. The following concepts, introduced earlier for protableaux, can be employed without changes already for tableaux: the concepts of a node *in use* and of a *useful* IC-tableau; the distinction between *P*-formulas of the *first kind* and the *second kind*, the definition of *irreducible* and *reducible nodes* of both kinds, the reduction of nodes of the first kind and the definition of a *1-irreducible* IC-tableau. Consider now the reduction process at a reducible node e of the second kind; the sets $\Gamma(e)$ and $\Delta(e)$ and the concepts connected with them are defined as before. Depending on the situation at the upper neighbour f of e , I now define *three* different reductions; the general frame remains the same, and the distinctions occur again at the essential base nodes r of $\Delta(e)$. The case 1 remains unchanged; in case 2 I start from the

Situation: f has a second lower neighbour g , and if g is of the second kind, then it is irreducible;

the reduction itself remains the same. The new reduction now occurs in

Case 3. Situation: f has a second lower neighbour g , and g is of the second kind and reducible.

Reduction: Below each r , insert a copy $\Delta_r(g)$ or $\Delta(g)$ (i.e. $\varepsilon'(r) = \varepsilon(g)$) on which t' copies t and, except at the essential base nodes, ε' copies ε . Further, copy all branches leaving $\Gamma(g)$ at a node which is either from the boundary or an unessential base node. Every essential base node q of $\Delta(g)$ is copied into a node q_r in $\Delta_r(g)$. Below each q_r , insert both $t(e)$, $t(g)$ on new lower neighbours q_{re} , q_{rg} of q_r (i.e. $\varepsilon'(q_r) = \varepsilon(f)$). Proceed below q_{re} as before below r and proceed below q_{rg} as before below q .

The verification that T' , t' , ε' is an IC-tableau is the same as in the case of protableaux; the arguments necessary in the new case 3 for the formula $t(g)$ are analogous to those carried out for $t(e)$ already. Also the final property of T' , t' , ε' , concerning the nodes k of the second kind in T' satisfying $k < f$, remains in effect in all its parts. Defining *irreducibility* for tableaux as we did before for protableaux, we can prove the

Reduction Theorem. *Every propositional IC-tableau T , t , ε can be transformed into an irreducible one for the same formula.*

For assume that T , t , ε is already useful and 1-irreducible. Define $M(T, t, \varepsilon)$ as before and let $N(T, t, \varepsilon)$ be the set of all nodes n of the second kind such that $e \leq n$ for some e in $M(T, t, \varepsilon)$; let $n(T, t, \varepsilon)$ be the number of elements in

$N(T, t, \varepsilon)$. If $N(T, t, \varepsilon)$ is empty, then also $M(T, t, \varepsilon)$ will be so; assume now that $M(T, t, \varepsilon)$ is not empty. Choose e in $M(T, t, \varepsilon)$ such that $\|e\|$ is maximal, and let T', t', ε' arise from the reduction process at e . In case 3 now g may belong to $M(T, t, \varepsilon)$, but it still follows from $\|g\| = \|e\|$ that there are no nodes from $M(T, t, \varepsilon)$ below g ; consequently, in T', t', ε' there are no reducible nodes k of the second kind such that $k < f$. The status of nodes h such that *not* $h \leq f$ has not been changed, but f itself may have been irreducible in T, t, ε and may be reducible in T', t', ε' . However, f was certainly in $N(T, t, \varepsilon)$, and thus $n(T', t', \varepsilon)$ is at most $n(T, t, \varepsilon) - 1$ in the cases 1 and 2 and at most $n(T, t, \varepsilon) - 2$ in the case 3. Repeating this reduction a finite number of steps, we arrive at an irreducible IC-tableau.

It is possible to achieve an even further going normalization, concerning Q -signed formulas $Qw_1 \wedge w_2$, $Qw_1 \vee w_2$. A node e with $t(e) = Qw_1 \wedge w_2$ is *irreducible* if it has one lower neighbour e' and e' has one lower neighbour e'' such that $\varepsilon(e) = e$, $\varepsilon(e') = e$, $t(e') = Qw_1$, $t(e'') = Qw_2$, and if there exists no h with $h < e'$ and $\varepsilon(h) = e$; it is obvious how to define a reduction achieving this irreducibility. A node e with $t(e) = Qw_1 \vee w_2$ is *irreducible* if it has two lower neighbours e' , e'' such that $\varepsilon(e) = e$, $t(e') = Qw_1$, $t(e'') = Qw_2$, and if there exists no h with $h < e$ and $\varepsilon(h) = e$. If e is not irreducible, we achieve irreducibility by inserting both e' , e'' and proceeding from them as was proceeded before from e ; every node h which was below e now will appear in two copies h_0 below e' and h_1 below e'' . If h was such that $\varepsilon(h) = e$, then the copy h_0 , say, will have two lower neighbours h'_0 , h''_0 with $t(h'_0) = Qw_1$, $t(h''_0) = Qw_2$; we then omit h'_0 as well as all nodes g with $g \leq h''_0$; for the copy h_1 we proceed analogously. In this manner, we can achieve that either (1) all nodes with $Q\wedge$ -formulas or (2) all nodes with $Q\vee$ -formulas become irreducible; we can, however, not achieve (1) and (2) simultaneously. Still, we can achieve (2) together with the following additional property of nodes e with $t(e) = Qw_1 \wedge w_2$: the component Qw_1 appears only at the lower neighbour of e ; the component Qw_2 appears on every branch through Qw_1 only once, and Qw_1 , Qw_2 are separated only by Q -formulas arising as components of $Q\vee$ -formulas. —If these reductions for Q -formulas are carried out before the reductions of P -formulas, then they remain preserved under the reductions of P -formulas and, therefore, may be obtained also for the irreducible IC-tableaux.

Irreducible IC-tableaux correspond to *irreducible LJ'-proofs* of sequents $\Rightarrow u$, characterized through the properties

- (p0) Every axiom $M, x \Rightarrow x$ occurs as premiss of a rule which employs the right-hand variable x as a side formula.
- (p1) If $\sigma : M \Rightarrow w$ is not the endsequent and if σ occurs as the conclusion of a $\neg$ -introduction ($w = \neg w_1$) or of the rule (I0 $\rightarrow$) ($w = w_1 \rightarrow w_2$), then the application of the next rule employs w as a side formula. Moreover, every application of (I1 $\rightarrow$) is, after some steps affecting only the left-hand side, followed by an application of (I0 $\rightarrow$) for the same principal formula; in

applications of $\neg$ -introduction and of $(I0\rightarrow)$ the side formula is not simultaneously parametric but actually disappears.

- (p2) If $\sigma : M \Rightarrow w$ is not the endsequent and if σ occurs as the conclusion of a $\wedge$ -introduction ($w = w_1 \wedge w_2$) or a $\vee$ -introduction ($w = w_1 \vee w_2$), then the application of the next rule employs w as a side formula.

Obviously, (p1) translates the irreducibility on nodes of the first kind, and (p0), (p2) translate the irreducibility of nodes of the second kind. It follows from the reduction theorem that every LJ'-proof of a sequent $\Rightarrow u$ can be transformed into an irreducible one.

4. Strategies and tableaux

This paragraph contains the final step in the proof of the equivalence theorem: it follows from 2.3 and 3.3 that the proof will be completed if we establish algorithms transforming formal E -strategies into (irreducible) IC-protableaux and vice versa.

4.1. From tableaux to strategies

Let T, t, ε be an irreducible IC-protableau for a formula u ; I shall define its *canonical extensions* S, δ, η . S is a tree, obtained from T by inserting new nodes and performing certain deformations; the function δ coincides with t on the nodes from T . *New nodes* are inserted as upper neighbours of the components of Q -signed $\wedge$ - and $\vee$ -formulas and of Q -signed quantified formulas, and δ assigns to them the appropriate symbolic attacks: if e has the one lower neighbour e' or the two lower neighbours e', e'' and

- if $t\varepsilon(e) = Qw_1 \wedge w_2$, $t(e') = Qw_i$, then $P\wedge_i$ is inserted below e ;
- if $t\varepsilon(e) = Qw_1 \vee w_2$, then $P\vee$ is inserted below e and above both e', e'' ;
- if $t\varepsilon(e) = Q\forall xw$, $t(e') = Qw(t)$, then Pt is inserted below e ,
- if $t\varepsilon(e) = Q\exists xw$, then $P\exists$ is inserted below e .

Deformations are performed below the nodes e such that $t\varepsilon(e) = Qw_1 \rightarrow w_2$, $t(e') = Pw_1$, $t(e'') = Qw_2$: in S the node e has only the one lower neighbour e' , whereas e' obtains, in addition to its lower neighbours from T , the new lower neighbour e'' (observe that, Pw_1 being irreducible, no deformation occurs immediately below e' ; still S may ramify threefold at e'). It should be noticed that the partial order of T is contained in the partial order of S ; for every node f of S I shall write W_f for the path defined by f in S ! — The function η is defined by assigning attacks to answers and formulas being attacked to attacks. The fact that answers determine attacks uniquely follows, for the new, P -signed attacks, from their definition; for the Q -signed attacks, present already in T, t, ε , it follows from (PT3) and from the irreducibility of nodes of the first kind.

Theorem. *The canonical extension of T, t, ε is a formal E -strategy.*

In the first, and principal, part of the proof, I shall prove by induction *in the tree* T the following two statements for every node e in T :

- (w0) The restrictions of δ, η to the S -path W_e form a formal E -dialogue.
- (w1) Let f be in S such that $f > e$. If f is Q -signed, then S does not ramify at f . If f is P -signed, then the lower neighbours of f in S represent precisely the possibilities required in a formal E -strategy.

Both statements are trivial for e_T . Assume that they hold for a node e which is not minimal in T ; I shall prove them for the lower neighbours e', e'' of e in T .

Case 1. e is P -signed. Then e is irreducible, and therefore e', e'' carry either Q -signed procomponents or, if $t(e)$ is $Pw_1 \rightarrow w_2$ or $P\neg w$, a Q -signed component of $t(e)$. This proves (w0). The statements in (w1) now have to be shown for e in place of f , and it follows from the definitions that the lower neighbours of e in S certainly represent all possibilities of Q -signed attacks. If there is also the possibility to proceed with an answer, i.e. if $t(e)$ itself is an attack, then e must be the lower neighbour g' of a node g such that $t\varepsilon(g) = Qw_1 \rightarrow w_2$, $t(e) = Pw_1$; the deformation below g then has the effect that in S the node e has a lower neighbour g'' representing the answer Qw_2 .

Case 20. e is Q -signed and $\varepsilon(e)$ is Q -signed. Then (w0), (w1) follow from the definition if $t\varepsilon(e)$ is $Q\neg w$; they follow from the way the new P -attacks were inserted if $t\varepsilon(e)$ is a quantified or an $\wedge$ - or a $\vee$ -formula; they follow from the deformation below e if $t\varepsilon(e)$ is $Qw_1 \rightarrow w_2$.

Case 21. e is Q -signed and $\varepsilon(e)$ is P -signed. Then $\varepsilon(e)$ is irreducible, and thus there exists a lower neighbour e_0 of $\varepsilon(e)$ with $e_0 \geq e$ such that e_0 carries a Q -signed attack upon $\varepsilon(e)$ which then is answered at the lower neighbour e' of e . In order to verify (D11), it has to be shown that no Q -signed attack started below e_0 is still open at e' . Such an attack would have to be directed upon a P -formula, and it follows from PT(2) that such a P -formula could not lie above $\varepsilon(e)$. But it follows again from PT(2) that there are no P -formulas on nodes f with $\varepsilon(e) > f \geq e$, and since our attack could not be directed upon $\varepsilon(e)$ itself we can conclude that it cannot exist. (Actually, we then can also conclude that there is no P -signed attack started below e_0 and still open at e' : such an attack could not itself be a P -formula and, therefore, would be one of the new, P -signed attacks which, by definition, are answered immediately.) This proves (w0), and (w1) again is evident.

In the second part of the proof it remains to be shown that P actually wins all the dialogues lying on the branches of S . Since T, t, ε is irreducible, all branches of T end with P -signed atomic formulas; thus all branches of S also end in this

way. Since atomic formulas cannot be attacked, the E -dialogues on the branches of S could be continued only if their last position is itself an attack which could be answered; such an attack would have to be upon a formula $Qw_1 \rightarrow w_2$. But the deformations performed in connection with these formulas show that the branches of S do not end with attacks Pw_1 . Thus the dialogues on these branches are won by P .

4.2. From strategies to tableaux

Let δ, η be a formal E -dialogue. A position n in $\text{def}(\delta)$ is said to be *virtual* if it carries an attack upon a $Q \rightarrow$ -formula which is answered (and then must be answered immediately). Remove from $\text{def}(\delta)$ all those positions which are either virtual or which carry a P -signed symbolic attack (i.e. an attack upon a quantified Q -formula or a $Q \wedge$ - or a $Q \vee$ -formula). Let E be the remaining subset of $\text{def}(\delta)$; the elements of E are called *nodes*. Both E and $\text{def}(\delta)$ are subsets of ω ; consider now E as well as $\text{def}(\delta)$ as linearly ordered sets, *ordered by the inverse* of the natural order of ω ; *only with respect to this inverse order* I shall now use the symbols $\leq$ and $<$ (thus e.g. $n+1 < n!$). I define a reference function ε on the set of non-minimal nodes e by $\varepsilon(e) = m$ if either $\eta(e+1) = [m, A]$ or if $\eta(e+1) = [n, D]$ and $\eta(n) = [m, A]$. The value $\varepsilon(e)$ belongs to E since symbolic attacks cannot be the objects of attacks themselves and since, in an E -dialogue, virtual positions cannot be attacked either. It follows from the definitions that ε describes the dissolution into procomponents and components of the formulas lying on the nodes of E , and this description is the same as that given by the reference function of a protableau on one of its branches. The condition (PT3) follows immediately from the rule (E); I shall now prove (PT2):

Lemma. *For every node e in $\text{def}(\varepsilon)$: if $\varepsilon(e)$ is P -signed, then $\varepsilon(e)$ is the smallest P -signed node f in E such that $e \leq f$.*

Assume first that e is P -signed. The P -formula $\delta(e)$ is either attacked or answered: since e is not virtual, $\delta(e)$ must be attacked. This attack takes place at $e+1$, implying $\varepsilon(e) = e$. Assume now that e is Q -signed. If $e+1$ carries an attack, then it is P -signed and, therefore, directed upon a Q -formula; in this case, therefore, $\varepsilon(e)$ is Q -signed. Consequently, if $\varepsilon(e)$ is P -signed, then $e+1$ must carry an answer to an attack upon $\varepsilon(e)$ which, therefore, takes place at $\varepsilon(e)+1$. It remains to be shown that, under these conditions, there are no P -signed nodes f such that $e < f < \varepsilon(e)$. Assume, to the contrary, that such nodes exist and let f be a maximal one; then $f < \varepsilon(e)+1$ since $\varepsilon(e)+1$ is Q -signed. I now shall examine all possibilities by which the P -formula $\delta(f)$ may have arisen, and I shall show that each of them leads to a contradiction.

Assume first that $\delta(f)$ is an attack. Then it must be directed upon a $Q \neg$ - or a $Q \rightarrow$ -formula; in the first case it will never be answered, and in the second case it cannot be answered since, in an E -dialogue, an answer would have to be made

immediately, causing f to become virtual. I thus arrive at a contradiction to (D11) since the attack at $\varepsilon(e)+1$ was made before the attack at f and is answered (at $e+1$). Assume now that $\delta(f)$ is an answer to an attack upon a P -formula at $\varepsilon(f)$; this attack then takes place at $\varepsilon(f)+1$. Since $e+1 \neq f$, it follows from (D12) that $\varepsilon(e)+1 \neq \varepsilon(f)+1$, hence $\varepsilon(e) \neq \varepsilon(f)$. Since $\varepsilon(e) \leq \varepsilon(f)$ due to the maximality of f , it follows that $\varepsilon(e) < \varepsilon(f)$ and, therefore, also $\varepsilon(e) < \varepsilon(f)+1$. Thus the attack at $\varepsilon(e)+1$ is later than the attack at $\varepsilon(f)+1$, and it follows from $e < f < \varepsilon(e)+1$ that this later attack is still open at f . I thus arrive at a contradiction to (D11) if the earlier attack shall be answered at f .

I shall now define the *canonical contraction* of a formal E -strategy S , δ , η . I begin by removing from S all nodes carrying P -signed symbolic attacks; let S' be the subtree of the remaining nodes. The tree T shall have the same nodes as S' ; its ordering arises from that of S' through certain deformations. Consider a node e' in S' carrying an attack Pw_1 upon a formula $Qw_1 \rightarrow w_2$. The upper neighbour e of e' in S is Q -signed and belongs to S' ; the tree S' does not ramify at e . In both S and S' there exists a lower neighbour e'' of e' carrying the answer to the attack at e' . The *deformation at e* re-defines the ordering such that both e' , e'' become lower neighbours of e ; I then proceed below e'' as before in S' , and I proceed below e' through those lower neighbours of e' in S' which are different from e'' ; if no such lower neighbours of e' exist then e' becomes minimal in T and will be said to be *conservational*. Beginning with a maximal node e' , I perform these deformations at all possible nodes e' and obtain the tree T . The function t shall be the restriction of δ to the nodes in T .

On every branch A of S there lies an E -dialogue which, according to the earlier discussion, determines a subset E_A of nodes; E_A is a subset of S' and a branch of T . The deformation at a node e' of S' has the effect that a branch A through e'' , containing e' only virtually, determines an E_A which does not contain e' . If $\delta(e')$ can be attacked, then the branches A containing such attacks determine branches E_A which contain e' . If, however, $\delta(e')$ is a P -signed atomic formula, then e' is conservational and belongs to the *conservational branch* $W_{e'}$ of T . It now follows that *every branch* of T is either a branch E_A or a conservational branch $W_{e'}$, obtained by cutting off some E_A at a suitable node e and adding the conservational node e' . Consequently, the *reference functions* ε_A defined earlier for every branch E_A , and therefore also for the conservational $W_{e'}$, coincide on the common parts of their domains. The function ε thus is defined as the union of the various functions ε_A . I call T , t , ε the *canonical contraction*.

Theorem. *The canonical contraction of S , δ , η is an irreducible IC-protaleau.*

That T , t , ε is an intuitionistic protaleau follows from the fact, observed earlier, that (PT2), (PT3) holds for every branch E_A . In order to see that this protaleau is intuitionistically closed, observe that the minimal nodes of the

branches W_e end with a P -signed atomic formula $\delta(e')$, and since W_e is part of a therefore, occur as the minimal nodes of the branches E_A ; it follows from (D10) that the branches E_A then are intuitionistically closed. But also the conservational branches W_e end with a P -signed atomic formula $\delta(e')$, and since W_e is part of a branch A of S , it follows again from (D10) that W_e is intuitionistically closed. Finally, the IC-protableau is irreducible since the branches A carry E -dialogues.

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